

Methods

1 Fourier series

1.1 Motivation

In the 1800s, Joseph Fourier was studying heat conduction along a metal rod, which led him to study 2π -periodic functions, that is $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(\theta + 2\pi) = f(\theta)$ for all $\theta \in \mathbb{R}$. He found that if

$$f(\theta) = \sum_n \hat{f}_n e^{in\theta},$$

then you can write down the coefficients $\{\hat{f}_n\}$ via the formula

$$\hat{f}_n = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta.$$

Actually, Fourier believed this worked for any 2π -periodic functions, that is if $f(\theta)$ is 2π -periodic and you compute each $\{\hat{f}_n\}$ and construct back the sum, it would return your original function. He was not quite correct.

1.2 Modern treatment

Let's introduce the vector space V of L -periodic functions, $V = \{f : \mathbb{R} \rightarrow \mathbb{C} \mid f \text{ "nice", } f(\theta + L) = f(\theta) \forall \theta \in \mathbb{R}\}$. Note for $f \in V$, we only need to consider values of f taken in an interval of length L , for example $[0, L)$ or $[-L/2, L/2)$ since periodicity covers elsewhere.

We can also introduce the inner product on V ,

Definition 1.1 – Inner product

$$\langle f, g \rangle = \int_0^L f(\theta) \overline{g(\theta)} d\theta$$

and the associated norm $\|f\| = \sqrt{\langle f, f \rangle}$. For $n \in \mathbb{Z}$, consider $e_n \in V$ defined by $e_n(\theta) = e^{2\pi i n \theta / L}$. Note that

$$\langle e_n, e_m \rangle = \int_0^L e^{2\pi i (n-m)\theta / L} d\theta = L \delta_{nm},$$

so $\{e_n\}$ are orthogonal and $\|e_n\|^2 = L$ for each n . Recall, if V_N is a N -dimensional vector space equipped with the usual inner (dot) product, and $\{\mathbf{e}_n\}_{n=1}^N$ are orthogonal with $\|\mathbf{e}_n\|^2 = L$, then for each $\mathbf{x} \in V_N$, we can write

$$\mathbf{x} = \sum_{n=1}^N \hat{x}_n \mathbf{e}_n$$

for some $\{\hat{x}_n\}$. To find $\{\hat{x}_n\}$, we could dot both sides with $\mathbf{e}_m$,

$$(\mathbf{x}, \mathbf{e}_m) = \sum_{n=1}^N \hat{x}_n (\mathbf{e}_n, \mathbf{e}_m) = \sum_{n=1}^N \hat{x}_n L \delta_{nm} = L \hat{x}_m$$

so $\hat{x}_n = (\mathbf{x}, \mathbf{e}_n)/L$. Could this work on V ? Granted, V is not finite dimensional (every subset of $\{e_n\}$ is linear independent), but we don't care. If we assume $f \in V$ can be written in the form $f(\theta) = \sum_n \hat{f}_n e_n(\theta)$, then

$$\langle f, e_m \rangle = \sum_n \hat{f}_n \langle e_n, e_m \rangle = \sum_n \hat{f}_n L \delta_{nm} = L \hat{f}_m$$

so

$$\hat{f}_n = \frac{1}{L} \langle f, e_n \rangle = \frac{1}{L} \int_0^L f(\theta) e^{-2\pi i n \theta / L} d\theta$$

which is the formula from before.

Definition 1.2 – Complex Fourier series

For L -periodic $f : \mathbb{R} \rightarrow \mathbb{C}$, its complex Fourier series is

$$\sum_n \hat{f}_n e^{2\pi i n \theta / L}$$

where

$$\hat{f}_n = \frac{1}{L} \int_0^L f(\theta) e^{-2\pi i n \theta / L} d\theta$$

are called the complex Fourier coefficients.

We will write, for $f \in V$,

$$f(\theta) \sim \sum_n \hat{f}_n e^{2\pi i n \theta / L}$$

to mean the series on the right corresponds to the complex Fourier series for the function of the left. We would like to replace $\sim$ with $=$ but this requires the next lecture.

If we split the complex Fourier series into $\{n = 0\} \cup \{n > 0\} \cup \{n < 0\}$,

$$\begin{aligned} \sum_n \hat{f}_n e^{2\pi i n \theta / L} &= \hat{f}_0 + \sum_{n=1}^{\infty} \hat{f}_n \left[\cos \frac{2\pi n \theta}{L} + i \sin \frac{2\pi n \theta}{L} \right] + \sum_{n=1}^{\infty} \hat{f}_{-n} \left[\cos \frac{2\pi n \theta}{L} - i \sin \frac{2\pi n \theta}{L} \right] \\ &= \hat{f}_0 + \sum_{n=1}^{\infty} \left[(\hat{f}_n + \hat{f}_{-n}) \cos \frac{2\pi n \theta}{L} + i(\hat{f}_n - \hat{f}_{-n}) \sin \frac{2\pi n \theta}{L} \right]. \end{aligned}$$

For $n \geq 0$, we define $a_n = \hat{f}_n + \hat{f}_{-n}$ and $b_n = i(\hat{f}_n - \hat{f}_{-n})$, and we get

Definition 1.3 – Fourier series

For L -periodic $f : \mathbb{R} \rightarrow \mathbb{C}$, its Fourier series is

$$\frac{1}{2} a_0 + \sum_{n=1}^{\infty} \left[a_n \cos \frac{2\pi n \theta}{L} + b_n \sin \frac{2\pi n \theta}{L} \right]$$

where

$$a_n = \frac{2}{L} \int_0^L f(\theta) \cos \frac{2\pi n \theta}{L} d\theta \quad \text{and} \quad b_n = \frac{2}{L} \int_0^L f(\theta) \sin \frac{2\pi n \theta}{L} d\theta$$

are called the Fourier coefficients.

If we set $c_n(\theta) = \cos 2\pi n \theta / L$ and $s_n(\theta) = \sin 2\pi n \theta / L$, we can show $\langle c_n, c_m \rangle = \langle s_n, s_m \rangle = L \delta_{nm} / 2$ and $\langle c_n, 1 \rangle = \langle s_n, 1 \rangle = \langle c_n, s_m \rangle = 0$, so $\{1, c_n, s_n\}_{n=1}^{\infty}$ is an orthogonal set in V .

Example 1.1

Take the 1-periodic $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(\theta) = \theta(1 - \theta)$ on $[0, 1)$. If we are taking the ordinary Fourier series, we expect all $b_n = 0$ since f is even. For $n \neq 0$,

$$\begin{aligned}\hat{f}_n &= \int_0^1 \theta(1 - \theta)e^{-2\pi i n \theta} d\theta \\ &= \left[\frac{-\theta(1 - \theta)e^{-2\pi i n \theta}}{2\pi i n} \right]_0^1 + \frac{1}{2\pi i n} \int_0^1 (1 - 2\theta)e^{-2\pi i n \theta} d\theta \\ &= \left[\frac{-(1 - 2\theta)e^{-2\pi i n \theta}}{(2\pi i n)^2} \right]_0^1 + 0 \\ &= -\frac{1}{2(\pi n)^2}\end{aligned}$$

and $\hat{f}_0 = 1/6$. So,

$$f(\theta) \sim \frac{1}{6} - \frac{1}{2} \sum_{n \neq 0} \frac{e^{2\pi i n \theta}}{(n\pi)^2} = \frac{1}{6} - \sum_{n=1}^{\infty} \frac{\cos 2\pi n \theta}{(n\pi)^2}.$$

1.3 Convergence of Fourier series

This subject is extremely subtle.

Definition 1.4 – Partial Fourier series

For L -periodic $f : \mathbb{R} \rightarrow \mathbb{C}$, we define the partial Fourier series by

$$(S_N f)(\theta) = \sum_{|n| \leq N} \hat{f}_n e^{2\pi i n \theta / L} = \frac{1}{2} a_0 + \sum_{n=1}^N \left[a_n \cos \frac{2\pi n \theta}{L} + b_n \sin \frac{2\pi n \theta}{L} \right].$$

It is natural to ask if $(S_N f) \rightarrow f$. However, we need to specify which mode of convergence we are considering. Is it pointwise, that is $(S_N f)(\theta) \rightarrow f(\theta)$ as $N \rightarrow \infty$ for each $\theta \in [0, L)$? Is it uniform, that is $\sup_{\theta \in [0, L)} |(S_N f)(\theta) - f(\theta)| \rightarrow 0$ as $N \rightarrow \infty$? Is it in norm, that is $\|S_N f - f\| = \sqrt{\int_0^L |(S_N f)(\theta) - f(\theta)|^2 d\theta} \rightarrow 0$ as $N \rightarrow \infty$? We will only consider pointwise convergence here.

Proposition 1.1

Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be a L -periodic function, for which on $[0, L)$

- (i) has finitely many discontinuities
- (ii) has finitely many local maxima and minima

then for each $\theta \in [0, L)$, we have

$$\frac{f(\theta_+) + f(\theta_-)}{2} = \lim_{N \rightarrow \infty} (S_N f)(\theta) = \sum_n \hat{f}_n e^{2\pi i n \theta / L}$$

where $f(\theta_{\pm}) = \lim_{\varepsilon \rightarrow 0} f(\theta \pm \varepsilon)$.

That is, at points of continuity, the Fourier series gives back the original function, and at points of discontinuity gives back the average of the function at the discontinuity. We call functions with properties (i) and (ii) Dirichlet functions. From now on, we assume all functions are Dirichlet functions so that $\sim$ means that the series on the right coincides with the function on the left at points of continuity and to its average at points of discontinuity. We will prove this proposition assuming f is smooth.

Proof. We assume $L = 2\pi$ and examine $\lim(S_N f)(\theta_0)$ for some $\theta_0 \in [0, 2\pi)$. By replacing $f(\theta)$ with $f(\theta + \theta_0)$, we can assume $\theta_0 = 0$ without loss of generality.

$$\begin{aligned}(S_N f)(0) &= \sum_{|n| \leq N} \hat{f}_n e^{in(0)} \\ &= \sum_{|n| \leq N} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta) e^{-in\theta} d\theta \right) \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta) \left[\sum_{|n| \leq N} e^{-in\theta} \right] d\theta\end{aligned}$$

The sum is a geometric series,

$$e^{iN\theta} \sum_{n=0}^{2N} e^{-in\theta} = \frac{\sin[(N + 1/2)\theta]}{\sin(\theta/2)}$$

if $\theta \in \mathbb{R} \setminus 2\pi\mathbb{Z}$ and sums to $2N + 1$ if $\theta \in 2\pi\mathbb{Z}$. So, we define the

Definition 1.5 – Dirichlet kernel

$$D_N(\theta) = \begin{cases} \sin[(N + 1/2)\theta] / \sin(\theta/2) & \text{if } \theta \in \mathbb{R} \setminus 2\pi\mathbb{Z} \\ 2N + 1 & \text{if } \theta \in 2\pi\mathbb{Z} \end{cases}$$

For each $N \geq 0$, D_N is continuous, even and 2π -periodic, also $\int_{-\pi}^{\pi} D_N(\theta) d\theta = 2\pi$. This means

$$f(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} D_N(\theta) f(0) d\theta,$$

so

$$\begin{aligned}(S_N f)(0) - f(0) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} D_N(\theta) [f(\theta) - f(0)] d\theta \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \sin \left[\left(N + \frac{1}{2} \right) \theta \right] F(\theta) d\theta\end{aligned}$$

where

$$F(\theta) = \frac{\theta}{\sin(\theta/2)} \left[\frac{f(\theta) - f(0)}{\theta} \right].$$

Note that $\theta \mapsto F(\theta)$ is smooth since

$$\begin{aligned}\frac{f(\theta) - f(0)}{\theta} &= \frac{1}{\theta} \int_0^{\theta} f'(t) dt \\ &= \frac{1}{\theta} \int_0^1 f'(\tau\theta) \theta d\tau \\ &= \int_0^1 f'(\tau\theta) d\tau.\end{aligned}$$

So, integrating by parts,

$$(S_N f)(0) - f(0) = \frac{1}{N + 1/2} \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos \left[\left(N + \frac{1}{2} \right) \theta \right] F'(\theta) d\theta \rightarrow 0 \text{ as } N \rightarrow \infty.$$

The proof gets harder when f is less regular. □

Example 1.2

Consider the 2π -periodic

$$f(\theta) = \begin{cases} 1 & \text{if } 0 \leq \theta < \pi \\ -1 & \text{if } -\pi \leq \theta < 0. \end{cases}$$

Since f is odd, $a_n = 0$ for each n and

$$\begin{aligned} b_n &= \frac{2}{2\pi} \int_{-\pi}^{\pi} f(\theta) \sin(n\theta) d\theta \\ &= \frac{2}{\pi} \int_0^{\pi} \sin(n\theta) d\theta \\ &= \frac{2}{n\pi} [1 - (-1)^n]. \end{aligned}$$

So,

$$f(\theta) \sim \frac{4}{\pi} \sum_{n \text{ odd}} \frac{\sin(n\theta)}{n}.$$

Note that the series is zero when $\theta = 0$, agreeing with the proposition at discontinuities.

1.4 Periodic extensions: cosine and sine series

Given $f : [0, L) \rightarrow \mathbb{C}$, we can define $2L$ -periodic even/odd extensions $f_{\text{even}}/f_{\text{odd}}$, where

$$f_{\text{even}}(\theta) = \begin{cases} f(\theta) & \text{if } \theta \in [0, L) \\ f(-\theta) & \text{if } \theta \in [-L, 0) \end{cases} \text{ and } f_{\text{odd}}(\theta) = \begin{cases} f(\theta) & \text{if } \theta \in [0, L) \\ -f(-\theta) & \text{if } \theta \in [-L, 0). \end{cases}$$

Note that $f(\theta) = f_{\text{even}}(\theta) = f_{\text{odd}}(\theta)$ if $\theta \in [0, L)$.

$$f_{\text{even}}(\theta) \sim \frac{1}{2} A_0 + \sum_{n=1}^{\infty} A_n \cos \frac{2\pi n\theta}{2L}$$

with

$$\begin{aligned} A_n &= \frac{2}{2L} \int_{-L}^L f_{\text{even}}(\theta) \cos \frac{2\pi n\theta}{2L} d\theta \\ &= \frac{2}{L} \int_0^L f(\theta) \cos \frac{n\pi\theta}{L} d\theta. \end{aligned}$$

And similarly,

$$f_{\text{odd}}(\theta) \sim \sum_{n=1}^{\infty} B_n \sin \frac{2\pi n\theta}{2L} \text{ with } B_n = \frac{2}{L} \int_0^L f(\theta) \sin \frac{n\pi\theta}{L} d\theta.$$

Definition 1.6 – Cosine and sine series

For $f : [0, L) \rightarrow \mathbb{C}$, we define its cosine series by

$$\frac{1}{2} A_0 + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi\theta}{L}$$

and its sine series by

$$\sum_{n=1}^{\infty} B_n \sin \frac{n\pi\theta}{L}$$

where A_n and B_n are defined as above.

Example 1.3

Take $f(\theta) = 1$ on $[0, \pi)$. For the sine series,

$$B_n = \frac{2}{\pi} \int_0^\pi \sin(n\theta) d\theta = \frac{2}{n\pi} [1 - (-1)^n].$$

On the interval $(0, \pi)$,

$$1 = f(\theta) = f_{\text{odd}}(\theta) = 4 \sum_{n \text{ odd}} \frac{\sin(n\theta)}{n\pi}.$$

For the cosine series, $A_0 = 2$ and $A_n = 0$ for $n \geq 1$. So, on $\theta \in [0, \pi)$, $1 = f(\theta) = f_{\text{even}}(\theta) = (1/2) \cdot 2 + 0$.

1.5 Regularity and the decay of Fourier coefficients**Lemma 1.1 – Riemann-Lebesgue lemma**

If $g : [a, b] \rightarrow \mathbb{C}$ is integrable on $[a, b]$ and $\lambda \in \mathbb{R}$, then

$$\int_a^b e^{-i\lambda\theta} g(\theta) d\theta \rightarrow 0 \text{ as } |\lambda| \rightarrow \infty.$$

So, if $f : \mathbb{R} \rightarrow \mathbb{C}$ is L -periodic and integrable on $[0, L)$, then

$$\hat{f}_n = \frac{1}{L} \int_0^L e^{-2\pi i n \theta / L} f(\theta) d\theta \rightarrow 0 \text{ as } |n| \rightarrow \infty.$$

Also, $a_n = \hat{f}_n + \hat{f}_{-n} \rightarrow 0$ and $b_n = i(\hat{f}_n - \hat{f}_{-n}) \rightarrow 0$ as $n \rightarrow \infty$.

Suppose f is L -periodic and $C^k(\mathbb{R})$, that is $f, f', \dots, f^{(k)}$ are continuous functions on $\mathbb{R}$.

$$\begin{aligned} \hat{f}_n &= \frac{1}{L} \int_0^L e^{-2\pi i n \theta / L} f(\theta) d\theta \\ &= \left[-\frac{1}{L} \frac{L}{2\pi i n} f(\theta) e^{-2\pi i n \theta / L} \right]_0^L + \frac{1}{L} \frac{L}{2\pi i n} \int_0^L e^{-2\pi i n \theta / L} f'(\theta) d\theta \\ &= -\frac{L}{2\pi i n} \frac{f(L^-) - f(0^+)}{L} + \frac{L}{2\pi i n} \frac{1}{L} \int_0^L e^{-2\pi i n \theta / L} f'(\theta) d\theta. \end{aligned}$$

By periodicity and continuity, $f(0^+) = f(L^+) = f(L^-)$, so the boundary term gives zero. Continuing this, we get

$$\hat{f}_n = \left(\frac{L}{2\pi i n} \right)^k \underbrace{\frac{1}{L} \int_0^L e^{-2\pi i n \theta / L} f^{(k)}(\theta) d\theta}_{o(1) \text{ by the R-L lemma}}.$$

Corollary 1.1

If f is $C^k(\mathbb{R})$, then

$$\hat{f}_n = o(1/n^k) \text{ as } |n| \rightarrow \infty.$$

1.6 Termwise differentiation

Suppose f is L -periodic and continuously differentiable on $[0, L)$ with $f' = g$ there. Then, g is continuous on $[0, L)$, so

$$\begin{aligned}\hat{g}_n &= \frac{1}{L} \int_0^L e^{-2\pi i n \theta / L} f'(\theta) d\theta \\ &= \frac{f(L^-) - f(0^+)}{L} + \frac{2\pi i n}{L} \frac{1}{L} \int_0^L e^{-2\pi i n \theta / L} f(\theta) d\theta.\end{aligned}$$

If f is continuous on $\mathbb{R}$, then by periodicity, $f(0^+) = f(L^+) = f(L^-)$, so $\hat{g}_n = 2\pi i n \hat{f}_n / L$, that is

$$f'(\theta) = g(\theta) \sim \sum_n \frac{2\pi i n}{L} \hat{f}_n e^{2\pi i n \theta / L} = \sum_n \frac{d}{d\theta} [\hat{f}_n e^{2\pi i n \theta / L}].$$

1.7 Parseval's theorem

If $f(\theta) \sim \sum_n \hat{f}_n e_n(\theta)$ and $g(\theta) \sim \sum_n \hat{g}_n e_n(\theta)$, then

$$\begin{aligned}\langle f, g \rangle &= \sum_{n,m} \hat{f}_n \overline{\hat{g}_m} \langle e_n, e_m \rangle \\ &= \sum_{n,m} \hat{f}_n \overline{\hat{g}_m} L \delta_{nm} \\ &= L \sum_n \hat{f}_n \overline{\hat{g}_n}.\end{aligned}$$

Theorem 1.1 – Parseval's theorem

$$\frac{1}{L} \int_0^L f(\theta) \overline{g(\theta)} d\theta = \sum_n \hat{f}_n \overline{\hat{g}_n}.$$

Or, as a special case,

$$\frac{1}{L} \int_0^L |f(\theta)|^2 d\theta = \sum_n |\hat{f}_n|^2.$$

Example 1.4

Take f to be 2π -periodic with $f(\theta) = \theta$ on $[-\pi, \pi)$. We can show that

$$f(\theta) \sim \sum_{n \neq 0} \frac{i(-1)^n}{n} e^{in\theta}.$$

By Parseval's theorem,

$$\begin{aligned}\frac{1}{2\pi} \int_{-\pi}^{\pi} \theta^2 d\theta &= \sum_{n \neq 0} \frac{1}{n^2} \\ \frac{\pi^2}{3} &= 2 \sum_{n=1}^{\infty} \frac{1}{n^2} \\ \frac{\pi^2}{6} &= \sum_{n=1}^{\infty} \frac{1}{n^2} = \zeta(2).\end{aligned}$$

It is also possible to show that $\zeta(2n) / \pi^{2n} \in \mathbb{Q}$.

2 Sturm-Liouville theory

2.1 Abstract eigenvalue problems

Recall from IA Vectors and Matrices, a linear map $A : V_N \rightarrow V_N$ was called Hermitian if $A^\dagger = A$, or equivalently $\mathbf{x} \cdot (A\mathbf{y}) = (A\mathbf{x}) \cdot \mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in V_N$. They had some nice properties.

- (i) Eigenvalues were real.
- (ii) Eigenvectors with distinct eigenvalues were orthogonal.
- (iii) We could pick an orthogonal set of eigenvectors $\{\mathbf{v}_i\}_{i=1}^N$ such that for each $\mathbf{x} \in V_N$,

$$\mathbf{x} = \sum_{i=1}^N \hat{x}_i \mathbf{v}_i \text{ where } \hat{x}_i = \frac{\mathbf{x} \cdot \mathbf{v}_i}{|\mathbf{v}_i|^2}.$$

What if $N = \infty$? We use the vector space of nice functions $f : [a, b] \rightarrow \mathbb{C}$ and we introduce the

Definition 2.1 – Inner product

$$\langle f, g \rangle_w = \int_a^b f(x) \overline{g(x)} w(x) dx,$$

where w is real valued and $w(x) > 0$ on (a, b) . We call w the weight function.

The associated norm is $\|f\|_w = \sqrt{\langle f, f \rangle_w}$. When $w = 1$, we will just write $\langle \cdot, \cdot \rangle$.

Definition 2.2 – Self-adjoint

A linear differential operator is said to be self-adjoint on $(V, \langle \cdot, \cdot \rangle_w)$ if

$$\langle Ly_1, y_2 \rangle_w = \langle y_1, Ly_2 \rangle_w$$

for all $y_1, y_2 \in V$.

Definition 2.3 – Eigenfunction and eigenvalue

We say $(y, \lambda) \in (V \setminus \{0\}) \times \mathbb{C}$ is an eigenfunction, eigenvalue pair for L if

$$Ly = \lambda y.$$

Proposition 2.1

If L is self-adjoint on $(V, \langle \cdot, \cdot \rangle_w)$, then

- (i) the eigenvalues are real;
- (ii) eigenfunctions with distinct eigenvalues are orthogonal with respect to $\langle \cdot, \cdot \rangle_w$;
- (iii) there exists a complete, orthogonal set of eigenfunctions $\{y_n\}_{n=1}^\infty$ that is for each $f \in V$, we can write

$$f = \sum_{n=1}^\infty \hat{f}_n y_n \text{ where } \hat{f}_n = \frac{\langle f, y_n \rangle_w}{\|y_n\|_w^2}.$$

Proof.

(i) If $Ly = \lambda y$ with $y \neq 0$,

$$\begin{aligned} (\lambda - \bar{\lambda}) \|y\|_w^2 &= \langle \lambda y, y \rangle_w - \langle y, \lambda y \rangle_w \\ &= \langle Ly, y \rangle_w - \langle y, Ly \rangle_w = 0 \end{aligned}$$

so $\lambda = \bar{\lambda}$.

(ii) If $Ly_1 = \lambda_1 y_1$ and $Ly_2 = \lambda_2 y_2$ with $y_1 \neq y_2$,

$$\begin{aligned} (\lambda_1 - \lambda_2) \langle y_1, y_2 \rangle_w &= \langle \lambda_1 y_1, y_2 \rangle_w - \langle y_1, \lambda_2 y_2 \rangle_w \\ &= \langle Ly_1, y_2 \rangle_w - \langle y_1, Ly_2 \rangle_w = 0 \end{aligned}$$

so $\langle y_1, y_2 \rangle_w = 0$.

(iii) We do a proof by authority, David Hilbert said it was true.

□

2.2 Self-adjoint operators and boundary values

We will study problems of the form $Ly = \lambda y$ with $a < x < b$ and y satisfying some boundary conditions at $x = a$ and $x = b$. L will be a particular type of differential operator.

Definition 2.4 – Sturm-Liouville operator

L is a Sturm-Liouville operator on (a, b) if it has the form

$$L = \frac{1}{w} \left[-\frac{d}{dx} \left(p \frac{d}{dx} \right) + q \right] = \frac{1}{w} \left[-p \frac{d^2}{dx^2} - p' \frac{d}{dx} + q \right]$$

where p, q, w are real-valued and $p, w > 0$ on (a, b) . We call w the weight function.

We see that $Ly = \lambda y$ is equivalent to

$$-\frac{d}{dx} \left(p \frac{dy}{dx} \right) + qy = \lambda wy$$

with $a < x < b$. We will enforce the boundary conditions by stipulating that y belongs to a suitable vector space of functions that have appropriate behaviour at the boundaries.

Definition 2.5 – Singular endpoint

For a Sturm-Liouville operator on (a, b) , we say an endpoint $c \in \{a, b\}$ is singular if

$$p(c) = 0$$

and non-singular otherwise.

We will impose real, homogeneous boundary conditions of the form

$$\alpha_c y(c) + \beta_c y'(c) = 0$$

where $\alpha_c, \beta_c \in \mathbb{R}$ and $\alpha_c^2 + \beta_c^2 \neq 0$ at each non-singular endpoint. We will work on generic vector spaces of the form

$$V = \{y \in C^2([a, b]) : y \text{ satisfies real, homogeneous boundary conditions at each non-singular endpoint}\}.$$

We will equip V with the inner product $\langle f, g \rangle_w = \int_a^b f(x) \overline{g(x)} w(x) dx$ where w is a weight function.

Example 2.1

$$-\frac{d}{dx} \left[\cos\left(\frac{x}{2}\right) \frac{dy}{dx} \right] + \sin\left(\frac{x}{2}\right) y = \lambda xy$$

with $0 < x < \pi$ and $y'(0) = 0$. $p(x) = \cos(x/2)$, $q(x) = \sin(x/2)$ and $w(x) = x$. We see that $x = \pi$ is singular, so the problem is of form $Ly = \lambda y$ with $y \in V$.

Example 2.2

$$-\frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] = \lambda y$$

with $-1 < x < 1$. $p(x) = 1-x^2$, $q(x) = 0$ and $w(x) = 1$. We see that $x = \pm 1$ are both singular, so we can simply take $V = \{y \in C^2([a, b])\}$ with $\langle f, g \rangle_w = \int_{-1}^1 f(x) \overline{g(x)} dx$.

Proposition 2.2

If L is a Sturm-Liouville operator on (a, b) with weight function w , then if $y_1, y_2 \in C^2([a, b])$, then

$$\langle Ly_1, y_2 \rangle_w - \langle y_1, Ly_2 \rangle_w = [p(x)(W(y_1, \overline{y_2}))(x)]_{x=a}^b$$

where W is the Wronskian $W(u, v) = uv' - vu'$.

So if $y_1, y_2 \in V$, then L is self-adjoint on $(V, \langle \cdot, \cdot \rangle_w)$.

Proof.

$$\begin{aligned} \langle Ly_1, y_2 \rangle_w - \langle y_1, Ly_2 \rangle_w &= \int_a^b \frac{1}{w} [-(py_1')' + qy_1] \overline{y_2} w dx - \int_a^b y_1 \frac{1}{w} [-(p\overline{y_2}')' + q\overline{y_2}] w dx \\ &= \int_a^b [y_1 (p\overline{y_2}')' - \overline{y_2} (py_1')'] dx \\ &= \int_a^b \frac{d}{dx} [p(x)(W(y_1, \overline{y_2}))(x)] dx \\ &= [p(x)(W(y_1, \overline{y_2}))(x)]_{x=a}^b. \end{aligned}$$

Now, assume $y_1, y_2 \in V$. If $x = c \in \{a, b\}$ is singular, then $p(c) = 0$, hence $[p(x)(W(y_1, \overline{y_2}))(x)]_{x=c} = 0$. If c is non-singular, then y_1, y_2 satisfy boundary conditions of the form $\alpha_c y(c) + \beta_c y'(c) = 0$ with $\alpha_c, \beta_c \in \mathbb{R}$ and $\alpha_c^2 + \beta_c^2 \neq 0$. Since $\alpha_c, \beta_c \in \mathbb{R}$, $\overline{y}$ also satisfies the same boundary conditions, hence

$$\begin{pmatrix} y_1(c) & y_1'(c) \\ \overline{y_2}(c) & \overline{y_2}'(c) \end{pmatrix} \begin{pmatrix} \alpha_c \\ \beta_c \end{pmatrix} = 0.$$

So,

$$0 = \begin{vmatrix} y_1(c) & y_1'(c) \\ \overline{y_2}(c) & \overline{y_2}'(c) \end{vmatrix} = (W(y_1, \overline{y_2}))(c)$$

and $[p(x)(W(y_1, \overline{y_2}))(x)]_{x=c} = 0$. Hence, $\langle Ly_1, y_2 \rangle_w - \langle y_1, Ly_2 \rangle_w = 0$ for all $y_1, y_2 \in V$. □

2.3 Sturm-Liouville eigenvalue problems

We were studying problems of the form

$$-\frac{d}{dx} \left[p \frac{dy}{dx} \right] + qy = \lambda wy$$

with $y \in V = \{y \in C^2([a, b]) : y \text{ satisfies real, homogeneous boundary conditions at each non-singular endpoint}\}$. We equipped V with the inner product $\langle \cdot, \cdot \rangle_w$. We can assume that elements of V are real-valued without loss of generality, since if $y = u + iv$ and $Ly = \lambda y$, then $Lu = \lambda u$ and $Lv = \lambda v$ since $p, q, w, \lambda \in \mathbb{R}$. So,

$$\langle y_1, y_2 \rangle_w = \int_a^b y_1(x) y_2(x) w(x) dx.$$

Since L is self-adjoint on $(V, \langle \cdot, \cdot \rangle_w)$, we know that there exists $(y_n, \lambda_n) \in (V \setminus \{0\}) \times \mathbb{R}$ such that $Ly_n = \lambda_n y_n$, $\langle y_n, y_m \rangle_w = 0$ if $\lambda_n \neq \lambda_m$, and for $f \in V$,

$$f(x) = \sum_{n=1}^{\infty} \hat{f}_n y_n(x)$$

with

$$\hat{f}_n = \frac{\langle f, y_n \rangle_w}{\|y_n\|_w^2}$$

being the generalised Fourier coefficients of f . It will also be the case that $\lambda_1 < \lambda_2 < \dots$ with $\lambda_n \rightarrow +\infty$ as $n \rightarrow \infty$.

Example 2.3

Consider

$$\begin{cases} -y'' = \lambda y \text{ on } 0 < x < L \\ y(0) = y(L) = 0 \end{cases}$$

so $p(x) = w(x) = 1$ and $q(x) = 0$ with $V = \{y \in C^2([0, L]) : y(0) = y(L) = 0\}$. We solve $y'' + \lambda y = 0$ so $y = A \sin(\sqrt{\lambda}x) + B \cos(\sqrt{\lambda}x)$. If $\lambda \leq 0$, we get invalid solutions, so $\lambda > 0$. $y(0) = 0$ requires $B = 0$. $y(L) = 0$ requires $A \sin(\sqrt{\lambda}L) = 0$, that is $\sqrt{\lambda} = n\pi/L$. Hence,

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2 \text{ and } y_n = \sin\left(\frac{n\pi x}{L}\right).$$

We see that $\lambda_n \rightarrow +\infty$ and

$$\langle y_n, y_m \rangle = \int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx = \frac{L}{2} \delta_{nm}.$$

For $f \in V$,

$$f(x) = \sum_{n=1}^{\infty} \hat{f}_n \sin\left(\frac{n\pi x}{L}\right) \text{ with } \hat{f}_n = \frac{\langle f, y_n \rangle}{\|y_n\|^2} = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

and we have re-derived the Fourier sine series.

2.4 Reduction to Sturm-Liouville form

Consider a general eigenvalue problem of the form

$$\alpha(x) \frac{d^2 y}{dx^2} + \beta(x) \frac{dy}{dx} + \gamma(x)y + \lambda y = 0$$

with $\alpha(x) > 0$. Dividing the equation by $\alpha(x)$ and multiplying by an integrating factor, we get that

Proposition 2.3

Any eigenvalue problem of the above form is equivalent to

$$-\frac{d}{dx} \left[p \frac{dy}{dx} \right] + qy = \lambda wy$$

where

$$p(x) = I(x) = \exp \left[\int^x \frac{\beta(t)}{\alpha(t)} dt \right]$$

$$q(x) = \frac{I(x)\gamma(x)}{\alpha(x)} \text{ and } w(x) = \frac{I(x)}{\alpha(x)}.$$

Proof.

$$-\frac{d}{dx} \left[p \frac{dy}{dx} \right] + qy - \lambda wy = I \left[-\frac{d^2 y}{dx^2} - \frac{\beta}{\alpha} \frac{dy}{dx} - \frac{\gamma}{\alpha} y - \frac{\lambda}{\alpha} y \right].$$

Since $I > 0$, we obtain the result. □

Example 2.4

Consider

$$\begin{cases} y'' - 2y' + \lambda y = 0 \text{ on } 0 < x < 1 \\ y(0) = y'(1) = 0. \end{cases}$$

We get that

$$I(x) = \exp \left[\int^x \frac{-2}{1} dx \right] = e^{-2x}$$

so the equation becomes

$$-\frac{d}{dx} \left[e^{-2x} \frac{dy}{dx} \right] = \lambda e^{-2x} y$$

and we see that the weight function is e^{-2x} . To solve this, we put $y \propto e^{\alpha x}$ to get $\alpha = 1 \pm \sqrt{1 - \lambda}$, that is if $\lambda \neq 1$, then the solutions are of the form

$$y = e^x [A' e^{x\sqrt{1-\lambda}} + B' e^{-x\sqrt{1-\lambda}}] = e^x [A \sin(\mu x) + B \cos(\mu x)]$$

where $1 - \lambda = -\mu^2$ with $\mu > 0$. We need $1 - \lambda < 0$ for non-trivial solutions. $y(0) = 0$ gives $B = 0$. $y'(1) = 0$ requires $Ae[\sin \mu + \mu \cos \mu] = 0$, that is $\tan \mu = -\mu$. By plotting a graph, we see that there is an infinite number of positive solutions $\mu_1, \mu_2, \dots$, so $\lambda_n = 1 + \mu_n^2$. Hence, $y_n(x) = e^x \sin(\mu_n x)$. We check that $\langle y_n, y_m \rangle_w \propto \delta_{nm}$,

$$\begin{aligned} \langle y_n, y_m \rangle_w &= \int_0^1 e^x \sin(\mu_n x) e^x \sin(\mu_m x) e^{-2x} dx \\ &= \frac{1}{2} \int_0^1 [\cos[(\mu_n - \mu_m)x] + \cos[(\mu_n + \mu_m)x]] dx \\ &= \frac{1}{2} \left[\frac{\sin(\mu_n - \mu_m)}{\mu_n - \mu_m} - \frac{\sin(\mu_n + \mu_m)}{\mu_n + \mu_m} \right] \\ &= \frac{1}{2} \left[\frac{\sin \mu_n \cos \mu_m - \sin \mu_m \cos \mu_n}{\mu_n - \mu_m} - \frac{\sin \mu_n \cos \mu_m + \sin \mu_m \cos \mu_n}{\mu_n + \mu_m} \right] \\ &= \frac{1}{2} \cos \mu_n \cos \mu_m \left[\frac{\tan \mu_n - \tan \mu_m}{\mu_n - \mu_m} - \frac{\tan \mu_n + \tan \mu_m}{\mu_n + \mu_m} \right] \\ &= \frac{1}{2} \cos \mu_n \cos \mu_m \left[\frac{-\mu_n + \mu_m}{\mu_n - \mu_m} - \frac{-\mu_n - \mu_m}{\mu_n + \mu_m} \right] \\ &= 0. \end{aligned}$$

2.5 Legendre's equation

Consider the eigenvalue problem

$$-\frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] = \lambda y$$

on $-1 < x < 1$. So $p = 1 - x^2$, $q = 0$ and $w = 1$. Since both endpoints are singular, we just need to work on $V = C^2([-1, 1])$. Since $x = 0$ is a regular point, we can look for solutions of the form $y = \sum_{n=0}^{\infty} a_n x^n$. Substituting this in, we get

$$a_{n+2} = \frac{n(n+1) - \lambda}{(n+1)(n+2)} a_n$$

and we obtain two linearly independent solutions:

$$y_0 = a_0 \left[1 + \frac{-\lambda}{2!} x^2 + \frac{-\lambda(6-\lambda)}{4!} x^4 + \dots \right] \text{ and } y_1 = a_1 \left[x + \frac{2-\lambda}{3!} x^3 + \frac{(2-\lambda)(12-\lambda)}{5!} x^5 + \dots \right].$$

Note that y_0 collapses if $\lambda = 0, 6, \dots$. Similarly, y_1 collapses if $\lambda = 2, 12, \dots$. In general, if $\lambda = k(k+1)$ with $k \in \mathbb{Z}_{\geq 0}$, either y_0 or y_1 becomes a polynomial. What if $\lambda \neq k(k+1)$? Since $|a_{n+2}/a_n| \rightarrow 1$, we know both series will converge on $|x| < 1$. What about $y(\pm 1)$? We only look at y_0 here (since y_1 is similar). Set $A_n = a_{2n}$, so

$$\frac{A_n}{A_{n+1}} = \frac{(2n+1)(2n+2)}{2n(2n+1) - \lambda} = 1 + \frac{1}{n} + \varepsilon_n$$

where $|\varepsilon_n| \leq M/n^2$ with $M = M(\lambda) > 0$. In particular, the right hand side is positive for sufficiently large n , say $n \geq N$. So $\{A_n\}$ will have the same sign for $n \geq N$. Using $e^x \geq 1 + x$, $|A_n|/|A_{n+1}| \leq e^{1/n} + |\varepsilon_n|$ so

$$|A_{n+1}| \geq \frac{e^{-1/n} |A_n|}{1 + e^{-1/n} |\varepsilon_n|} \geq \frac{e^{-1/n} |A_n|}{1 + |\varepsilon_n|} \geq e^{-1/n} |A_n| e^{-|\varepsilon_n|}.$$

Hence, for $n \geq N$,

$$|A_{n+1}| \geq |A_N| \exp \left[- \left(\frac{1}{n} + \frac{1}{n-1} + \dots + \frac{1}{N} \right) - (|\varepsilon_n| + \dots + |\varepsilon_N|) \right] \geq |A_N| e^{-H_n + H_{N-1}} e^{-M\pi^2/6}.$$

Since $H_n \leq \log n + 2\gamma$,

$$|A_{n+1}| \geq |A_N| e^{H_{N-1} - M\pi^2/6} e^{-\log n - 2\gamma} > \frac{C}{n+1}$$

for $n \geq N$. Moreover, we can write y_0 as

$$y_0 = \sum_{n \leq N} A_n x^{2n} + \sum_{n > N} A_n x^{2n}.$$

Since $\{A_n\}$ have the same sign, we can assume that they are positive without loss of generality. Note

$$\sum_{n > N} A_n x^{2n} > C \left[\sum_{n=1}^{\infty} \frac{x^{2n}}{n} - \sum_{n \leq N} \frac{x^{2n}}{n} \right] = C \left[\log \left(\frac{1}{1-x^2} \right) - \text{some polynomial} \right] \rightarrow +\infty$$

as $x \rightarrow \pm 1$, that is $y_0 \notin V$ in general. So we must have $\lambda_k = k(k+1)$. This gives an even polynomial of degree k from y_0 and an odd polynomial of degree k from y_1 . With the normalisation $y(1) = 1$, the solutions are called Legendre polynomials. We list the first few Legendre polynomials here.

$$\begin{aligned} P_0(x) &= 1 \\ P_1(x) &= x \\ P_2(x) &= (3x^2 - 1)/2 \\ P_3(x) &= (5x^3 - 3x)/2 \end{aligned}$$

We can also show that

$$\int_{-1}^1 P_n(x) P_m(x) dx = \frac{2\delta_{nm}}{2n+1}.$$

2.6 Bessel's equation

Fix an integer $m \geq 0$ and consider

$$-\frac{d}{dr} \left[r \frac{dy}{dr} \right] + \frac{m^2}{r} y = \lambda r y$$

with $0 < r < 1$ and $y(1) = 0$. We see that $p = r$, $q = m^2/r$ and $w = r$. Expanding out the derivatives,

$$r^2 y'' + r y' + (\lambda r^2 - m^2) y = 0.$$

Set $z = \sqrt{\lambda} r$ (λ is unknown, but we can show that $\lambda > 0$), and set $R(z) = y(z)$. We then get

$$z^2 R'' + z R' + (z^2 - m^2) R = 0$$

with $0 < z < \sqrt{\lambda}$ and $R(\sqrt{\lambda}) = 0$. This is Bessel's equation of order m . Since $z = 0$ is a regular singular point, we can get solutions of the form $z^\sigma \sum_{n=0}^{\infty} a_n z^n$. We then obtain two linearly independent solutions, only one of which is non-singular as $z \rightarrow 0$. Label the corresponding solution $R = J_m(z)$ with

$$J_m(z) = \left(\frac{z}{2}\right)^m \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+m)!} \left(\frac{z}{2}\right)^{2n}.$$

These are called the Bessel functions of the first kind of order m . We can also show that

$$J_m(z) = \sqrt{\frac{2}{\pi z}} \cos\left(z - \frac{\pi m}{2} - \frac{\pi}{4}\right) + \mathcal{O}\left(\frac{1}{z^{3/2}}\right)$$

as $z \rightarrow \infty$, so $J_m(z)$ has infinitely many zeros on the positive z -axis and we label them j_{mk} . Since we require $J_m(\sqrt{\lambda}) = 0$, that is $\lambda = \lambda_k = j_{mk}^2$, the solutions to the original equation are

$$y_k(r) = J_m(j_{mk} r)$$

with $\lambda_k = j_{mk}^2$. Also, $\langle y_k, y_l \rangle_w \propto \delta_{kl}$.

2.7 Inhomogeneous problems

Let L be a Sturm-Liouville operator. Consider problems of the form $Ly = f$ with $y, f \in V$. Without loss of generality, we can take $w = 1$. Let $\{y_k\}$ be the normalised eigenfunctions of L . By completeness, we can write $y = \sum A_k y_k$ and $f = \sum B_k y_k$. So,

$$\sum_{k=1}^{\infty} (\lambda_k A_k - B_k) y_k = 0$$

which implies $\lambda_k A_k = B_k$ by orthogonality. So, if $\lambda_k \neq 0$, $A_k = B_k / \lambda_k$. Then,

$$y(x) = \sum_{k=1}^{\infty} \frac{B_k}{\lambda_k} y_k(x) \text{ with } B_k = \langle f, y_k \rangle.$$

Hence,

$$y(x) = \sum_{k=1}^{\infty} \frac{1}{\lambda_k} \left[\int_a^b f(\xi) y_k(\xi) d\xi \right] y_k(x) = \int_a^b G(x; \xi) f(\xi) d\xi$$

where

$$G(x; \xi) = \sum_{k=1}^{\infty} \frac{y_k(\xi) y_k(x)}{\lambda_k}$$

are called Green's functions.

3 Linear PDEs and separation of variables

3.1 A reminder on superposition

We will be interested in solving boundary value problems (BVPs) of the form

$$\begin{cases} P\psi(\mathbf{x}) = 0 & \mathbf{x} \in \Omega \\ \text{some BCs} & \mathbf{x} \in \partial\Omega \end{cases}$$

and initial boundary value problems (IBVPs) of the form

$$\begin{cases} Q\varphi(\mathbf{x}, t) = 0 & (\mathbf{x}, t) \in \Omega \times (0, T) \\ \text{some ICs} & (\mathbf{x}, t) \in \Omega \times \{t = 0\} \\ \text{some BCs} & (\mathbf{x}, t) \in \partial\Omega \times (0, T) \end{cases}$$

where P and Q are linear partial differential operators and Ω is a bounded, open subset of $\mathbb{R}^n$. Here, BCs mean boundary conditions and ICs mean initial conditions.

We can split the IBVP into

$$\begin{cases} Q\varphi_1(\mathbf{x}, t) = 0 & Q\varphi_2(\mathbf{x}, t) = 0 & (\mathbf{x}, t) \in \Omega \times (0, T) \\ \text{ICs} = 0 & \text{some ICs} & (\mathbf{x}, t) \in \Omega \times \{t = 0\} \\ \text{some BCs} & \text{BCs} = 0 & (\mathbf{x}, t) \in \partial\Omega \times (0, T) \end{cases}$$

and we see that $\varphi = \varphi_1 + \varphi_2$ will solve the original IBVP. In this course, Ω will always be (possibly after a change of variables) a line/rectangle/cuboid. So, without loss of generality, we can always deal with BCs that are zero everywhere apart from on one endpoint/edge/face. For example, consider

$$\begin{cases} P\psi(\mathbf{x}) = 0 & \mathbf{x} \in (0, 1) \times (0, 1) \\ \psi = f_i & \text{on side } i \text{ with } i = 1, 2, 3, 4. \end{cases}$$

We could look at four problems for $\{\psi_i\}_{i=1}^4$ with

$$\begin{cases} P\psi_i(\mathbf{x}) = 0 & \mathbf{x} \in (0, 1) \times (0, 1) \\ \psi_i = 0 & \text{on side } \neq i \\ \psi_i = f_i & \text{on side } i. \end{cases}$$

Then, $\psi = \psi_1 + \psi_2 + \psi_3 + \psi_4$ solves the original problem.

3.2 Laplace's equation

Recall, for $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$, Laplace's equation is

$$\Delta\varphi = 0$$

where $\Delta = \nabla \cdot \nabla$. For example, on $\mathbb{R}^3$ and in Cartesian coordinates, $\Delta = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$. We say φ is harmonic if $\Delta\varphi = 0$. Harmonic functions are always infinitely differentiable. An example from physics, if $\mathbf{u} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is the velocity of an incompressible fluid ($\nabla \cdot \mathbf{u} = 0$) that is irrotational ($\nabla \times \mathbf{u} = \mathbf{0}$), then we can solve via Laplace's equation. Since $\nabla \times \mathbf{u} = \mathbf{0}$ on $\mathbb{R}^3$, there exists $\varphi : \mathbb{R}^3 \rightarrow \mathbb{R}$ such that $\mathbf{u} = \nabla\varphi$. Then, $0 = \nabla \cdot \mathbf{u} = \nabla \cdot (\nabla\varphi) = \Delta\varphi$.

We will consider BVPs of the form

$$\begin{cases} \Delta\varphi(\mathbf{x}) = 0 & \mathbf{x} \in \Omega \\ B\varphi = f(\mathbf{x}) & \mathbf{x} \in \partial\Omega \end{cases}$$

where $B\varphi = \varphi$ (Dirichlet) or $B\varphi = \partial\varphi/\partial\mathbf{n}$ (Neumann), or even $B\varphi = \varphi + \partial\varphi/\partial\mathbf{n}$ (Robin).

3.2.1 Separation of variables on a square

Consider

$$\begin{cases} \varphi_{xx} + \varphi_{yy} = 0 & (x, y) \in (0, 1) \times (0, 1) \\ \varphi(x, y) = 0 & \text{on } x = 0, x = 1, y = 0 \\ \varphi(x, 1) = f(x). \end{cases}$$

We try a separable solution of the form

$$\varphi = X(x)Y(y).$$

Then, $X''(x)Y(y) + X(x)Y''(y) = 0$. Dividing through by XY ,

$$\frac{X''(x)}{X(x)} + \frac{Y''(y)}{Y(y)} = 0.$$

So, there exists $\lambda \in \mathbb{R}$ such that

$$\frac{X''}{X} = -\lambda \text{ and } \frac{Y''}{Y} = \lambda.$$

We first look at X equation, since $\varphi = 0$ at $x = 0, 1$, we have

$$\begin{cases} X'' + \lambda X = 0 & \text{on } 0 < x < 1 \\ X(0) = 0 \\ X(1) = 0. \end{cases}$$

This is a Sturm-Liouville problem, so there exists $(X_n, \lambda_n)_{n=1}^{\infty}$ such that $\lambda_1 < \lambda_2 < \dots$ and $\langle X_n, X_m \rangle \propto \delta_{nm}$. We can check that $\lambda > 0$ is required for non-trivial solutions. The general solution is then $X(x) = A \sin(\sqrt{\lambda}x) + B \cos(\sqrt{\lambda}x)$. The boundary condition $X(0) = 0$ gives $B = 0$ and $X(1) = 0$ gives $A \sin(\sqrt{\lambda}) = 0$ so $\lambda_n = n^2\pi^2$. Combining, we get

$$X_n(x) = \sin(n\pi x)$$

with $\lambda_n = n^2\pi^2$ and we know from before that $\langle X_n, X_m \rangle = \delta_{nm}/2$.

The Y problem becomes $Y'' - n^2\pi^2 Y = 0$ with $Y(0) = 0$. We find $Y = A \sinh(n\pi y) + B \cosh(n\pi y)$. $Y(0) = 0$ gives $B = 0$, so

$$Y_n(y) = A_n \sinh(n\pi y).$$

Hence, we have found functions $\{\varphi_n\}_{n=1}^{\infty}$ with

$$\varphi_n(x, y) = A_n \sin(n\pi x) \sinh(n\pi y),$$

each satisfying $\Delta\varphi_n = 0$ in $(0, 1) \times (0, 1)$ and $\varphi_n = 0$ on $x = 0, x = 1, y = 0$. As the problem is linear, the same is true for

$$\varphi(x, y) = \sum_{n=1}^{\infty} A_n \sin(n\pi x) \sinh(n\pi y).$$

We want $\varphi(x, 1) = f(x)$, that is

$$f(x) = \sum_{n=1}^{\infty} A_n \sin(n\pi x) \sinh(n\pi).$$

By orthogonality,

$$\langle f, X_n \rangle = \sum_{m=1}^{\infty} A_m \langle X_m, X_n \rangle \sinh(m\pi) = \frac{A_n}{2} \sinh(n\pi).$$

That is, the solution is

$$\varphi(x, y) = \sum_{n=1}^{\infty} \frac{2\langle f, X_n \rangle}{\sinh(n\pi)} \sin(n\pi x) \sinh(n\pi y).$$

3.2.2 Separation of variables on a disc or annulus

We want to solve $\Delta\varphi = 0$ in the region $r_1 \leq |x| \leq r_2$ in the (x, y) plane. We use polar coordinates (r, θ) so that Laplace's equation becomes

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \varphi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \varphi}{\partial \theta^2} = 0.$$

Once again, we look for separable solutions of the form $\varphi = R(r)\Theta(\theta)$, and we get

$$\frac{r(rR')'}{R} + \frac{\Theta''}{\Theta} = 0$$

so there exists $\lambda \in \mathbb{R}$ such that

$$r(rR')' = \lambda R \text{ and } \Theta'' + \lambda \Theta = 0.$$

The solution to the Θ equation is

$$\begin{cases} A \cos(\sqrt{\lambda}\theta) + B \sin(\sqrt{\lambda}\theta) & \text{if } \lambda > 0 \\ A\theta + B & \text{if } \lambda = 0 \\ A \cosh(\sqrt{|\lambda|}\theta) + B \sinh(\sqrt{|\lambda|}\theta) & \text{if } \lambda < 0. \end{cases}$$

However, we need $\Theta(\theta + 2\pi) = \Theta(\theta)$ for all θ . So, we require $\lambda = n^2$ with $n \in \mathbb{Z}_{\geq 0}$, and we get

$$\begin{cases} \Theta_0(\theta) = A \\ \Theta_n(\theta) = A_n \cos(n\theta) + B_n \sin(n\theta). \end{cases}$$

Plugging $\lambda = n^2$ into the R equation, $r(rR')' = n^2 R$. For $n = 0$, we find $R_0(r) = A + B \log r$. For $n > 0$, we try $R(r) = r^\alpha$ and get $\alpha^2 = n^2$, that is $\alpha = \pm n$. So, $R_n(r) = A_n r^n + B_n r^{-n}$.

Combining, we get the very general solution

$$\varphi(r, \theta) = A + B \log r + \sum_{n=1}^{\infty} [A_n r^n + B_n r^{-n}] [C_n \cos(n\theta) + D_n \sin(n\theta)].$$

If the point $r = 0$ belongs to our domain, then we must take $B = B_n = 0$ for all n .

Example 3.1

Solve

$$\begin{cases} \Delta\varphi = 0 & r_1 < r < r_2 \\ \varphi = 0 & r = r_1 \\ \varphi = f(\theta) & r = r_2. \end{cases}$$

We repeat the same analysis, but we require $R_n(r_1) = 0$ for all n . Then,

$$\begin{cases} R_0(r) = \log(r/r_1) \\ R_n(r) = (r/r_1)^n - (r_1/r)^n \end{cases}$$

and we get the solution of the form

$$\varphi(r, \theta) = C_0 \log \left(\frac{r}{r_1} \right) + \sum_{n=1}^{\infty} \left[\left(\frac{r}{r_1} \right)^n - \left(\frac{r_1}{r} \right)^n \right] [A_n \cos(n\theta) + B_n \sin(n\theta)].$$

The boundary condition on $r = r_2$ now reads

$$f(\theta) = C_0 \log \left(\frac{r_2}{r_1} \right) + \sum_{n=1}^{\infty} \left[\left(\frac{r_2}{r_1} \right)^n - \left(\frac{r_1}{r_2} \right)^n \right] [A_n \cos(n\theta) + B_n \sin(n\theta)]$$

which is of the form $\frac{1}{2}a_0 + \sum_{n=1}^{\infty}[a_n \cos(n\theta) + b_n \sin(n\theta)]$. That is, the right hand side should be the Fourier series for f , so

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \cos(n\theta) d\theta \text{ and } b_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \sin(n\theta) d\theta.$$

For a more general problem,

$$\begin{cases} \Delta\varphi = 0 & r_1 < r < r_2 \\ \varphi = g(\theta) & r = r_1 \\ \varphi = f(\theta) & r = r_2. \end{cases}$$

We can simply use superposition and write $\varphi = \varphi_1 + \varphi_2$ where $\varphi_i = 0$ on $r = r_i$.

3.2.3 Separation of variables on a ball or shell

We want to solve $\Delta\varphi = 0$ in the region $a < |\mathbf{x}| < b$ in $\mathbb{R}^3$, but we assume the problem is symmetric about the z -axis. That is, if (r, θ, ϕ) are the usual spherical polars, then $\varphi = \varphi(r, \theta)$. In spherical polars, Laplace's equation is

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \frac{\partial \varphi}{\partial r} \right] + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left[\sin \theta \frac{\partial \varphi}{\partial \theta} \right] + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \varphi}{\partial \phi^2} = 0.$$

From the symmetry assumed, $\partial^2 \varphi / \partial \phi^2 = 0$. We look for separable solutions of the form $\varphi = R(r)\Theta(\theta)$, then

$$\frac{[r^2 R']'}{R} + \frac{1}{\Theta \sin \theta} [\sin \theta \Theta']' = 0$$

so there must exist $\lambda \in \mathbb{R}$ such that

$$[r^2 R']' = \lambda R \text{ and } -[\sin \theta \Theta']' = \lambda \sin \theta \Theta$$

with $0 \leq \theta \leq \pi$.

In the Θ equation, use $x = \cos \theta$ and set $\tilde{\Theta}(x) = \Theta(\theta)$. Note

$$\frac{d\Theta}{d\theta} = \frac{dx}{d\theta} \frac{d\tilde{\Theta}}{dx} = -\sin \theta \frac{d\tilde{\Theta}}{dx}$$

so $d/d\theta = -\sin \theta d/dx$. So the equation becomes

$$-\left(-\sin \theta \frac{d}{dx}\right) \left[\sin \theta \left(-\sin \theta \frac{d\tilde{\Theta}}{dx}\right) \right] = \lambda \sin \theta \tilde{\Theta}.$$

Since $\sin^2 \theta = 1 - x^2$,

$$-\sin \theta \frac{d}{dx} \left[(1 - x^2) \frac{d\tilde{\Theta}}{dx} \right] = \lambda \sin \theta \tilde{\Theta}.$$

So for $0 < \theta < \pi$, that is $-1 < x < 1$, we get

$$-\frac{d}{dx} \left[(1 - x^2) \frac{d\tilde{\Theta}}{dx} \right] = \lambda \tilde{\Theta}$$

for $-1 < x < 1$. This is Legendre's equation, and we know $\tilde{\Theta}(x) = P_n(x)$ with $\lambda = \lambda_n = n(n+1)$. That is, for $n \geq 0$,

$$\Theta_n(\theta) = P_n(\cos \theta) \text{ with } \lambda_n = n(n+1).$$

The R equation now becomes $[r^2 R']' = n(n+1)R$ and we try $R = r^\alpha$ to get $\alpha(\alpha+1) = n(n+1)$. That is, $\alpha = n$ or $\alpha = -(n+1)$.

Combining, we get the general solution

$$\varphi(r, \theta) = \sum_{n=0}^{\infty} \left(\alpha_n r^n + \frac{\beta_n}{r^{n+1}} \right) P_n(\cos \theta).$$

If $\{r = 0\}$ belongs to our domain, we would need $\beta_n = 0$ for all n .

Example 3.2

Consider

$$\begin{cases} \Delta \varphi = 0 & \text{on } r < 1 \\ \varphi(r, \theta) = f(\theta) & \text{on } r = 1 \end{cases}$$

so

$$\varphi(r, \theta) = \sum_{n=0}^{\infty} \alpha_n r^n P_n(\cos \theta).$$

The boundary condition now reads

$$f(\theta) = \sum_{n=0}^{\infty} \alpha_n P_n(\cos \theta).$$

Recall from example sheets that

$$\langle P_n, P_m \rangle = \int_{-1}^1 P_n(x) P_m(x) dx = \frac{2\delta_{nm}}{2n+1}.$$

Equivalently, setting $x = \cos \theta$,

$$\frac{2\delta_{nm}}{2n+1} = \int_0^\pi P_n(\cos \theta) P_m(\cos \theta) \sin \theta d\theta.$$

So, multiplying the boundary condition by $\sin \theta P_m(\cos \theta)$ then integrating between $\theta \in (0, \pi)$, we get

$$\int_0^\pi f(\theta) P_m(\cos \theta) \sin \theta d\theta = \sum_{n=0}^{\infty} \alpha_n \langle P_n, P_m \rangle = \frac{2\alpha_m}{2m+1},$$

that is

$$\alpha_n = \frac{2n+1}{2} \int_0^\pi f(\theta) P_n(\cos \theta) \sin \theta d\theta.$$

3.3 Wave equation

Consider a piece of taut string under constant tension τ , clamped at its ends $x = 0$ and $x = L$. Let $y(x, t)$ denote the vertical displacement of the string. Assume the oscillations are transverse and the slope $|y_x| \ll 1$. Note that

$$|y(x, t)| = \left| \int_0^x y_x(s, t) ds \right| \leq \int_0^L |y_x| ds \ll 1$$

so the displacement is small as well. Consider two points x_A and x_B close on the string, with $\delta x = x_B - x_A$ and x being the midpoint. The string makes an angle θ_A with the horizontal at A and θ_B at B . Since the oscillations are transverse, we can resolve the forces horizontally, so $\tau \cos \theta_B - \tau \cos \theta_A = 0$. The mass of the small bit of string is proportional to its length, that is

$$\int_{x_A}^{x_B} ds = \int_{x_A}^{x_B} \sqrt{1 + y_x^2} dx \approx \delta x.$$

It is reasonable to assume the mass to be $\mu \delta x$, where $\mu > 0$ is the density per unit length. Resolving vertically and using Newton's second law,

$$\mu \delta x \frac{\partial^2 y}{\partial t^2} = \tau \sin \theta_B - \tau \sin \theta_A.$$

Dividing by $\tau \approx \tau \cos \theta_A \approx \tau \cos \theta_B$, we get

$$\frac{\mu}{\tau} \delta x \frac{\partial^2 y}{\partial t^2} = \frac{\tau \sin \theta_B}{\tau \cos \theta_B} - \frac{\tau \sin \theta_A}{\tau \cos \theta_A} = \left[\frac{\partial y}{\partial x} \right]_B - \left[\frac{\partial y}{\partial x} \right]_A.$$

By the mean value theorem,

$$\frac{\mu}{\tau} \delta x \frac{\partial^2 y}{\partial t^2} = \frac{\partial^2 y}{\partial x^2} \delta x.$$

Dividing by δx and taking $\delta x \rightarrow 0$, we arrive at the wave equation

$$\frac{1}{c^2} \frac{\partial^2 y}{\partial t^2} = \frac{\partial^2 y}{\partial x^2}$$

where $c = \sqrt{\tau/\mu}$ is the wave speed. We have the boundary conditions $y(0, t) = y(L, t) = 0$ and the initial conditions $y(x, 0) = f(x)$ and $y_t(x, 0) = g(x)$.

3.3.1 Waves on a string

We want to solve the IBVP

$$\begin{cases} y_{tt} - c^2 y_{xx} = 0 & (x, t) \in (0, L) \times (0, \infty) \\ y(0, t) = y(L, t) = 0 & t \in (0, \infty) \\ y(x, 0) = f(x) & x \in (0, L) \\ y_t(x, 0) = g(x) & x \in (0, L). \end{cases}$$

We do separation of variables and try $y = X(x)T(t)$ with $X(0) = X(L) = 0$. Using dots to denote differentiation with respect to time, we get

$$\frac{\ddot{T}}{c^2 T} = \frac{X''}{X}$$

so there exists $\lambda \in \mathbb{R}$ such that

$$\begin{cases} X'' + \lambda X = 0 \\ X(0) = X(L) = 0 \end{cases} \quad \text{and} \quad \ddot{T} + \lambda c^2 T = 0.$$

The X equation is a Sturm-Liouville problem and the solutions are $X_n(x) = \sin(n\pi x/L)$ with $\lambda_n = (n\pi/L)^2$. The T equation becomes $\ddot{T} + (n\pi c/L)^2 T = 0$ which yields $T_n(t) = A_n \cos(n\pi c t/L) + B_n \sin(n\pi c t/L)$. By superposition,

$$y(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right) \left[A_n \cos\left(\frac{n\pi c t}{L}\right) + B_n \sin\left(\frac{n\pi c t}{L}\right) \right]$$

solves the wave equation with $y(0, t) = y(L, t) = 0$.

The initial conditions become

$$f(x) = \sum_{n=1}^{\infty} A(n) \sin\left(\frac{n\pi x}{L}\right) \quad \text{and} \quad g(x) = \sum_{n=1}^{\infty} \frac{n\pi c}{L} B_n \sin\left(\frac{n\pi x}{L}\right)$$

which is the sine series. By orthogonality $\langle X_n, X_m \rangle = (L/2)\delta_{nm}$, we get

$$A_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \quad \text{and} \quad B_n = \frac{L}{n\pi c} \frac{2}{L} \int_0^L g(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

3.3.2 Waves on a drum

The higher dimension analogue of the wave equation is

$$\frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = \Delta \varphi.$$

We want to solve the IBVP on a drum $\Omega = \{(r, \theta) : 0 \leq r < 1, 0 \leq \theta < 2\pi\}$.

$$\begin{cases} \varphi_{tt} - c^2 \Delta \varphi = 0 & \Omega \times (0, \infty) \\ \varphi = 0 & \partial\Omega \times (0, \infty) \\ \varphi = f & \Omega \times \{t = 0\} \\ \varphi_t = g & \Omega \times \{t = 0\}. \end{cases}$$

For simplicity, assume $f = f(r)$ and $g = g(r)$, that is $\varphi = \varphi(r, t)$. With these assumptions, the wave equation is

$$\frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \varphi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \varphi}{\partial \theta^2}.$$

We try solutions of the form $\varphi = R(r)T(t)$, so

$$\frac{\ddot{T}}{c^2 T} = \frac{[rR']'}{rR}.$$

There exists $\lambda \in \mathbb{R}$ such that

$$\begin{cases} -\frac{d}{dr} \left[r \frac{dR}{dr} \right] = \lambda r R & 0 < r < 1 \\ R(1) = 0 \end{cases} \quad \text{and} \quad \ddot{T} + c^2 \lambda T = 0.$$

The R equation is the same as the Bessel problem with $m = 0$. The solutions then are $R_k(r) = J_0(j_{0k}r)$ with $\lambda_k = j_{0k}^2$. The T equation becomes $\ddot{T} + (cj_{0k})^2 T = 0$, that is $T_k(t) = A_k \cos(j_{0k}ct) + B_k \sin(j_{0k}ct)$. So,

$$\varphi(r, t) = \sum_{k=1}^{\infty} J_0(j_{0k}r) [A_k \cos(j_{0k}ct) + B_k \sin(j_{0k}ct)]$$

solves the wave equation and the boundary conditions.

The initial conditions are

$$f(r) = \sum_{k=1}^{\infty} A_k J_0(j_{0k}r) \quad \text{and} \quad g(r) = \sum_{k=1}^{\infty} B_k j_{0k} c J_0(j_{0k}r).$$

Recall that

$$\langle R_k, R_l \rangle_r = \int_0^1 J_0(j_{0k}r) J_0(j_{0l}r) r dr = \frac{1}{2} J_0'(j_{0k})^2 \delta_{kl}.$$

So, by orthogonality,

$$A_k = \frac{2}{J_0'(j_{0k})^2} \int_0^1 f(r) J_0(j_{0k}r) r dr \quad \text{and} \quad B_k = \frac{2}{J_0'(j_{0k})^2} \frac{1}{cj_{0k}} \int_0^1 g(r) J_0(j_{0k}r) r dr.$$

3.3.3 Energy conservation and uniqueness

Consider a string under tension τ with density μ clamped between $x = 0$ and $x = L$. Define the energy

$$E(t) = \int_0^L \left[\frac{1}{2} \mu y_t^2 + \frac{1}{2} \tau y_x^2 \right] dx.$$

The first term is kinetic energy and the second term is the potential energy stored in the string, which is proportional to the extension of the string. On a small patch, this is $ds - dx = (\sqrt{1 + y_x^2} - 1)dx \approx y_x^2/2 dx$. For simplicity, set $\mu = 1$ so $\tau = c^2$ and

$$E(t) = \frac{1}{2} \int_0^L [y_t^2 + c^2 y_x^2] dx.$$

For the wave equation on a general $\Omega \times (0, \infty)$, we define the energy as

$$E(t) = \frac{1}{2} \int_{\Omega} [\varphi_t^2 + c^2 |\nabla \varphi|^2] dV.$$

In the drum example before, the potential energy is again proportional to the amount of stretch in the drum skin. $dS - dx dy = \sqrt{1 + |\nabla \varphi|^2} dx dy - dx dy \approx |\nabla \varphi|^2/2 dx dy$.

Recall our IBVP from before. We look at the evolution of energy as φ evolves according to the IBVP.

$$\begin{aligned} \frac{dE}{dt} &= \int_{\Omega} [\varphi_t \varphi_{tt} + c^2 \nabla \varphi \cdot \nabla \varphi_t] dV \\ &= c^2 \int_{\Omega} [\varphi_t \Delta \varphi + \nabla \varphi \cdot \nabla \varphi_t] dV \\ &= c^2 \int_{\Omega} \nabla \cdot (\varphi_t \nabla \varphi) dV \\ &= c^2 \int_{\partial \Omega} \varphi_t \frac{\partial \varphi}{\partial \mathbf{n}} dS. \end{aligned}$$

We know that $\varphi(\mathbf{x}, t) = 0$ for $\mathbf{x} \in \partial \Omega$ so $\varphi_t(\mathbf{x}, t) = 0$ on $\partial \Omega$. Hence, $dE/dt = 0$ and so $E(t) = E(0)$. From initial conditions, we conclude that

$$E(t) = E(0) = \frac{1}{2} \int_{\Omega} [g^2 + c^2 |\nabla f|^2] dV$$

and energy is conserved.

Proposition 3.1

The solution to

$$\begin{cases} \varphi_{tt} - c^2 \Delta \varphi = 0 & \Omega \times (0, \infty) \\ \varphi = 0 & \partial \Omega \times (0, \infty) \\ \varphi = f & \Omega \times \{t = 0\} \\ \varphi_t = g & \Omega \times \{t = 0\} \end{cases}$$

is unique.

Proof. Suppose φ_1, φ_2 satisfies the IBVP. Consider $\psi = \varphi_1 - \varphi_2$, so ψ satisfies the IBVP with $f = g = 0$. By energy conservation, $E(t) = E(0) = 0$ so

$$0 = E(t) = \frac{1}{2} \int_{\Omega} [\psi_t^2 + c^2 |\nabla \psi|^2] dV.$$

That is, $\psi_t = |\nabla \psi| = 0$ and so ψ is constant in Ω . Since $\psi = 0$ on $\partial \Omega$, $\psi = 0$ and so $\varphi_1 = \varphi_2$. $\square$

3.3.4 Wave reflection and transmission

Consider an infinite string under constant tension τ with density μ_+ on $x > 0$ and density μ_- on $x < 0$. The equations of motion are

$$\begin{cases} \frac{1}{c_-^2} \frac{\partial^2 y}{\partial t^2} = \frac{\partial^2 y}{\partial x^2} & \text{on } x < 0 \\ \frac{1}{c_+^2} \frac{\partial^2 y}{\partial t^2} = \frac{\partial^2 y}{\partial x^2} & \text{on } x > 0 \end{cases}$$

where $c_{\pm}^2 = \tau/\mu_{\pm}$. Consider, for $\omega \in \mathbb{R}$, an incoming wave from $x < 0$ of the form

$$y(x, t) = \cos[\omega(t - x/c_-)] = \operatorname{Re}[e^{i\omega(t-x/c_-)}].$$

We expect some part of the wave to get transmitted and some to get reflected due to the interactions at $x = 0$. The string would not break so

$$y(0^+, t) = y(0^-, t).$$

Since the point at $x = 0$ has no mass, the forces must match there. Resolving vertically, $\tau \sin \theta_+ = \tau \sin \theta_-$ so

$$\frac{\partial y}{\partial x}(0^+, t) = \frac{\partial y}{\partial x}(0^-, t).$$

We consider solutions of the form

$$y(x, t) = \begin{cases} e^{i\omega(t-x/c_-)} + Re^{i\omega(t+x/c_-)} & \text{on } x < 0 \\ Te^{i\omega(t-x/c_+)} & \text{on } x > 0. \end{cases}$$

Imposing the conditions on $x = 0$, we have

$$e^{i\omega t} + Re^{i\omega t} = Te^{i\omega t} \text{ and } \frac{i\omega}{c_-}e^{i\omega t}(R-1) = -\frac{i\omega}{c_+}Te^{i\omega t}.$$

Writing $\lambda = c_+/c_- = \sqrt{\mu_-/\mu_+}$, we get

$$R = \frac{\lambda - 1}{\lambda + 1} \text{ and } T = \frac{2\lambda}{\lambda + 1}$$

where R is the reflection coefficient and T is the transmission coefficient. If $\mu_+ = \mu_-$, we get $\lambda = 1$, $R = 0$ and $T = 1$, which is just a normal string. If $\mu_+ \gg \mu_-$, we get $\lambda \ll 1$, $R \approx -1$ and $T \approx 0$. Nearly none of the wave gets transmitted and we have a near complete reflection with a phase change of π . If $\mu_- \gg \mu_+$, we get $\lambda \gg 1$, $R \approx 1$ and $T \approx 2$.

3.4 The heat equation

The temperature of some conductive material $\varphi(\mathbf{x}, t)$ satisfies the heat equation

$$\frac{\partial \varphi}{\partial t} = \kappa \Delta \varphi$$

where κ is a positive constant. We are interested in the IBVP

$$\begin{cases} \varphi_t - \kappa \Delta \varphi = 0 & \Omega \times (0, \infty) \\ \varphi = 0 & \partial\Omega \times (0, \infty) \\ \varphi = f & \Omega \times \{t = 0\}. \end{cases}$$

We try solutions of the form $\varphi = T(t)\psi(\mathbf{x})$. Plugging in, we find there exists $\mu \in \mathbb{R}$ such that

$$\dot{T} + \kappa\mu T = 0 \text{ and } -\Delta\psi = \mu\psi.$$

The T equation yields $T(t) = e^{-\kappa\mu t}$. Considering the boundary conditions, we want to solve

$$\begin{cases} -\Delta\psi = \mu\psi & \mathbf{x} \in \Omega \\ \psi = 0 & \mathbf{x} \in \partial\Omega. \end{cases}$$

The solutions to this depend on the geometry of Ω .

3.4.1 Heat conduction on a square sheet

Take $\Omega = \{(x, y) \in (0, L) \times (0, L)\}$. Again, try $\psi = X(x)Y(y)$ and we get

$$\frac{X''}{X} + \frac{Y''}{Y} + \mu = 0.$$

So, there exists $\lambda \in \mathbb{R}$ such that $X'' + \lambda X = 0$ and $Y'' + (\mu - \lambda)Y = 0$. The boundary conditions require $X(0) = X(L) = 0$ and $Y(0) = Y(L) = 0$.

The X equation gives $X_n(x) = \sin(n\pi x/L)$ with $\lambda_n = (n\pi/L)^2$. So, $Y = A \sin(y\sqrt{\mu - \lambda_n}) + B \cos(y\sqrt{\mu - \lambda_n})$. $Y(0) = 0$ implies $B = 0$ and $Y(L) = 0$ gives $\mu_{mn} - \lambda_n = (m\pi/L)^2$. Hence, $Y_n(y) = \sin(m\pi y/L)$ with $\mu_{mn} = (n\pi/L)^2 + (m\pi/L)^2$. By superposition,

$$\varphi(x, y, t) = \sum_{m,n=1}^{\infty} A_{mn} e^{-\kappa \mu_{mn} t} \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi y}{L}\right)$$

satisfies $\varphi_t - \kappa \Delta \varphi = 0$ and $\varphi = 0$ on $\partial\Omega$.

The initial condition reads

$$f(x, y) = \sum_{m,n=1}^{\infty} A_{mn} \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi y}{L}\right).$$

Using $\langle X_n, X_k \rangle = L\delta_{nk}/2$ and other orthogonality relations, we get

$$A_{mn} = \frac{2}{L} \frac{2}{L} \int_0^L dx \int_0^L dy f(x, y) \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi y}{L}\right).$$

3.4.2 Heat flow down a pipe

In cylindrical polars, $\Omega = \{(\rho, \phi, z) : 0 \leq \rho < 1, 0 \leq \phi < 2\pi, 0 < z < L\}$. For simplicity, assume $f = f(\rho, z)$ is independent of ϕ and we look for solutions of the form $\varphi = T(t)\psi(\rho, z)$. We get

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \psi}{\partial \rho} \right) + \frac{\partial^2 \psi}{\partial z^2} + \mu \psi = 0$$

and we try $\psi(\rho, z) = P(\rho)Z(z)$. So, there exists $\lambda \in \mathbb{R}$ such that

$$-\frac{1}{\rho P} [\rho P']' = \lambda \text{ and } Z'' + (\mu - \lambda)Z = 0.$$

That is,

$$-\frac{d}{d\rho} \left[\rho \frac{dP}{d\rho} \right] = \lambda \rho P$$

for $0 < \rho < 1$ and $P(1) = 0$. This is Bessel's problem with $m = 0$, so the solution is $P_k(\rho) = J_0(j_{0k}\rho)$ with $\lambda_k = j_{0k}^2$. The Z equation becomes $Z'' - (\mu - \lambda_k)Z = 0$ with $Z(0) = Z(L) = 0$. Hence, $Z_n(z) = \sin(n\pi z/L)$ with $\mu_{kn} - \lambda_k = (n\pi/L)^2$. By superposition,

$$\varphi(\rho, z, t) = \sum_{n,k=1}^{\infty} B_{nk} e^{-\kappa \mu_{kn} t} J_0(j_{0k}\rho) \sin\left(\frac{n\pi z}{L}\right)$$

satisfies $\varphi_t - \kappa \Delta \varphi = 0$ and $\varphi = 0$ on $\partial\Omega$.

The initial condition is

$$f(\rho, z) = \sum_{n,k=1}^{\infty} B_{nk} J_0(j_{0k}\rho) \sin\left(\frac{n\pi z}{L}\right).$$

Recall that

$$\int_0^1 J_0(j_{0k}\rho) J_0(j_{0l}\rho) \rho d\rho = \frac{1}{2} J_0'(j_{0k})^2 \delta_{kl}$$

and we get

$$B_{nk} = \frac{2}{J_0'(j_{0k})^2} \frac{2}{L} \int_0^1 d\rho \int_0^L dz \rho f(\rho, z) J_0(j_{0k}\rho) \sin\left(\frac{n\pi z}{L}\right).$$

3.4.3 Heat loss and uniqueness

For a solution to the IBVP, define the heat energy as

$$Q(t) = \frac{1}{2} \int_{\Omega} \varphi(\mathbf{x}, t)^2 dV.$$

Then,

$$\begin{aligned} Q'(t) &= \int_{\Omega} \varphi_t \varphi dV \\ &= \int_{\Omega} \kappa \varphi \Delta \varphi dV \\ &= \kappa \int_{\Omega} [\nabla \cdot (\varphi \nabla \varphi) - |\nabla \varphi|^2] dV \\ &= \kappa \underbrace{\int_{\partial\Omega} \varphi \frac{\partial \varphi}{\partial \mathbf{n}} dS}_{= 0 \text{ by BC}} - \kappa \underbrace{\int_{\Omega} |\nabla \varphi|^2 dV}_{\geq 0} \end{aligned}$$

so $Q'(t) \leq 0$. Hence, $Q(t) \leq Q(0) = \frac{1}{2} \int_{\Omega} f(\mathbf{x})^2 dV$.

Proposition 3.2

The solution to

$$\begin{cases} \varphi_t - \kappa \Delta \varphi = 0 & \Omega \times (0, \infty) \\ \varphi = 0 & \partial\Omega \times (0, \infty) \\ \varphi = f & \Omega \times \{t = 0\} \end{cases}$$

is unique.

Proof. Suppose φ_1, φ_2 satisfy the IBVP and set $\varphi = \varphi_1 - \varphi_2$. Then, φ satisfies the IBVP with $f = 0$.

$$\frac{1}{2} \int_{\Omega} \varphi(\mathbf{x}, t)^2 dV = Q(t) \leq Q(0) = 0$$

so $\varphi(\mathbf{x}, t) = 0$ on $\Omega \times (0, \infty)$, that is $\varphi_1 = \varphi_2$. □

However, this might not always work as $Q(t)$ and the integral might not exist for certain φ .

4 Inhomogeneous problems and Green's functions

4.1 The Dirac delta function

We might have seen the mysterious delta function $\delta(x)$ with the following properties. $\delta(x) = 0$ for all $x \neq 0$, and

$$\int_{-\varepsilon}^{\varepsilon} \delta(x) dx = 1$$

for all $\varepsilon > 0$. How do we make this more rigorous? We can define a sequence of functions

$$\delta_n(x) = \begin{cases} (n/\alpha) \exp[-1/(1-n^2x^2)] & \text{for } |x| < 1/n \\ 0 & \text{for } |x| \geq 1/n \end{cases} \quad \text{where } \alpha = \int_{-1}^1 \exp\left[-\frac{1}{1-y^2}\right] dy.$$

Then, $\delta_n(x)$ is smooth for each n . Also, for all $\varepsilon > 0$, there exists N such that $\int_{-\varepsilon}^{\varepsilon} \delta_n(x) dx = 1$ for all $n > N$. To see this, given any $\varepsilon > 0$, take $N = 1/\varepsilon$. So, if $n > N$, then $1/n < \varepsilon$ and

$$\begin{aligned} \int_{-\varepsilon}^{\varepsilon} \delta_n(x) dx &= \int_{-1/n}^{1/n} \delta_n(x) dx \\ &= \frac{1}{\alpha} \int_{-1}^1 \exp\left[-\frac{1}{1-y^2}\right] dy \\ &= 1 \end{aligned}$$

where $y = nx$. We see that $\lim_{n \rightarrow \infty} \delta_n(x) = 0$ if $x \neq 0$ and for all $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \int_{-\varepsilon}^{\varepsilon} \delta_n(x) dx = 1.$$

It almost looks like $\delta(x) = \lim_{n \rightarrow \infty} \delta_n(x)$, but the pointwise limit does not exist everywhere since

$$\lim_{n \rightarrow \infty} \delta_n(0) = \lim_{n \rightarrow \infty} n/(\alpha e) = \infty.$$

However, the limit does exist in a weaker sense. The delta function rarely appears in isolation, for example,

$$\int_{-\infty}^{\infty} f(x) \delta(x-a) dx = f(a)$$

or $\ddot{y} + \omega^2 y = \delta(t)$. We can interpret these as sequences of statements that we will eventually take the limit of. So,

$$\int_{-\infty}^{\infty} f(x) \delta(x-a) dx = \lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f(x) \delta_n(x-a) dx$$

or solving $\ddot{y}_n + \omega^2 y_n = \delta_n(t)$ and defining $y(t) = \lim_{n \rightarrow \infty} y_n(t)$. We could even look $\delta'(x)$ via

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) \delta'(x) dx &= \lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f(x) \delta'_n(x) dx \\ &= - \lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f'(x) \delta_n(x) dx \\ &= - \int_{-\infty}^{\infty} f'(x) \delta(x) dx \\ &= -f'(0). \end{aligned}$$

It gets irritating writing all these limits, so we stop doing it and treat $\delta(x)$ like an ordinary function.

4.1.1 Periodic delta functions

Let $\delta^L(x)$ denote the L -periodic extension of $\delta(x)$ outside $[-L/2, L/2]$. It has Fourier coefficients

$$\frac{1}{L} \int_{-L/2}^{L/2} \delta(x) e^{-2\pi i n x / L} dx = \frac{1}{L}$$

so

$$\delta^L(x) \sim \frac{1}{L} \sum_n e^{2\pi i n x / L}.$$

The right hand side does not even converge, but is this plausible?

$$\begin{aligned}
f(0) &= \int_{-L/2}^{L/2} f(x) \delta^L(x) dx \\
&= \sum_n \left[\frac{1}{L} \int_{-L/2}^{L/2} f(x) e^{2\pi i n x / L} dx \right] \\
&= \sum_n \hat{f}_{-n} \\
&= \sum_n \hat{f}_n \\
&= \sum_n \hat{f}_n e^{2\pi i n(0)/L} = f(0)
\end{aligned}$$

which is good. Note that the Fourier series for each $\delta_n^L(x)$ converge rapidly.

4.1.2 Eigenfunction expansion of the delta function

Let L be a Sturm-Liouville operator on V , where V are all the twice-differentiable functions on $[a, b]$ that satisfy real, homogeneous boundary conditions at each non-singular endpoint. Let $\{Y_k\}_{k=1}^\infty$ be the normalised eigenfunctions. Fix $\xi \in (a, b)$ and consider the functions $x \mapsto \delta_n(x - \xi)$. For n sufficiently large, $\delta_n(x - \xi) \in V$. By completeness,

$$\delta_n(x - \xi) = \sum_{k=1}^\infty \langle \delta_n(\cdot - \xi), Y_k \rangle_w Y_k(x) = \sum_{k=1}^\infty \left[\int_a^b \delta_n(t - \xi) Y_k(t) w(t) dt \right] Y_k(x).$$

Formally, letting $n \rightarrow \infty$,

$$\delta(x - \xi) = \sum_{k=1}^\infty w(\xi) Y_k(\xi) Y_k(x).$$

Is it plausible? For $f \in V$,

$$\begin{aligned}
f(x) &= \int_a^b f(\xi) \delta(x - \xi) d\xi \\
&= \sum_{k=1}^\infty \left[\int_a^b f(\xi) Y_k(\xi) w(\xi) d\xi \right] Y_k(x) \\
&= \sum_{k=1}^\infty \hat{f}_k Y_k(x)
\end{aligned}$$

where $\hat{f}_k = \langle f, Y_k \rangle_w$, which is exactly the eigenfunction expansion.

4.2 Green's functions

We want to solve the inhomogeneous problem

$$\begin{cases} Ly = f(x) & a < x < b \\ y(a) = 0 \\ y(b) = 0 \end{cases}$$

where $L = \alpha(x) d^2/dx^2 + \beta(x) d/dx + \gamma(x)$ with $\alpha(x) \neq 0$ on (a, b) . Fix $\xi \in (a, b)$. Suppose we can find a function $G = G(x, \xi)$ such that

$$\begin{cases} L[G(x, \xi)] = \delta(x - \xi) & a < x < b \\ G(a, \xi) = 0 \\ G(b, \xi) = 0. \end{cases}$$

Consider

$$y(x) = \int_a^b G(x, \xi) f(\xi) d\xi.$$

Clearly $y(a) = y(b) = 0$ and

$$Ly = \int_a^b L[G(x, \xi)] f(\xi) d\xi = \int_a^b \delta(x - \xi) f(\xi) d\xi = f(x)$$

so $y(x)$ solves the problem. We call G the Green's function for L with Dirichlet boundary conditions.

Note that on $a < x < \xi$ and on $\xi < x < b$, G satisfies $L[G(x, \xi)] = 0$. This is a nice, second-order, homogeneous ODE, so we expect $x \mapsto G(x, \xi)$ to be well-behaved. In the neighbourhood of $x = \xi$, the worst term on the left is $\alpha(x) d^2G/dx^2$, and the worst term on the right is $\delta(x - \xi)$. So we expect

$$\alpha(x) \frac{d}{dx} \left[\frac{dG}{dx} \right] = \delta(x - \xi) - \text{more regular terms.}$$

What do we need for $d/dx[\text{something}] \sim \delta(x - \xi)$? Recall the sequence $\{\delta_n\}$ were such that

$$\lim_{n \rightarrow \infty} \delta_n(x) = 0 \text{ for } x \neq 0 \text{ and } \lim_{n \rightarrow \infty} \int_{-\varepsilon}^{\varepsilon} \delta_n(x) dx = 1 \text{ for all } \varepsilon > 0.$$

It is natural to look at

$$H_n(x) = \int_{-\infty}^x \delta_n(t) dt$$

since $H'_n(x) = \delta_n(x)$. If we fix $x > 0$, then for n sufficiently large ($1/n < x$), we have

$$H_n(x) = \int_{-1/n}^{1/n} \delta_n(t) dt = 1.$$

Similarly, if we fix $x < 0$, then for $-1/n > x$, we have $H_n(x) = 0$. So, we can define the Heaviside function as

$$\lim_{n \rightarrow \infty} H_n(x) = H(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$$

and $H'(x - \xi) = \delta(x - \xi)$. Hence, we expect to need dG/dx to behave like $H(x - \xi)$ near $x = \xi$ and so we expect a jump discontinuity at $x = \xi$. We see

$$G(x, \xi) = \int_a^x \frac{dG(t, \xi)}{dt} dt$$

and

$$\int_a^x H(t - \xi) dt = \begin{cases} x - \xi & \text{if } x \geq \xi \\ 0 & \text{if } x < \xi \end{cases}$$

which is continuous. Hence, to conclude, we expect $x \mapsto G(x, \xi)$ to be continuous at $x = \xi$ and $x \mapsto dG(x, \xi)/dx$ to have a jump discontinuity at $x = \xi$.

To fix the jump discontinuity, integrate

$$\frac{d^2G}{dx^2} + \frac{\beta(x)}{\alpha(x)} \frac{dG}{dx} + \frac{\gamma(x)}{\alpha(x)} G = \frac{\delta(x - \xi)}{\alpha(x)}$$

over $(\xi - \varepsilon, \xi + \varepsilon)$ with $\varepsilon > 0$ to get

$$\left[\frac{dG}{dx} \right]_{\xi - \varepsilon}^{\xi + \varepsilon} + \int_{\xi - \varepsilon}^{\xi + \varepsilon} \left[\frac{\beta}{\alpha} \frac{dG}{dx} + \frac{\gamma}{\alpha} G \right] dx = \int_{\xi - \varepsilon}^{\xi + \varepsilon} \frac{\delta(x - \xi)}{\alpha(x)} dx = \frac{1}{\alpha(\xi)}.$$

Taking $\varepsilon \rightarrow 0$, we get

$$\left[\frac{dG}{dx} \right]_{\xi -}^{\xi +} = \frac{1}{\alpha(\xi)}.$$

In summary, G satisfies

$$G(\xi^+, \xi) = G(\xi^-, \xi) \text{ and } \left[\frac{dG}{dx} \right]_{\xi^-}^{\xi^+} = \frac{1}{\alpha(x)}.$$

Example 4.1

Let $L = d^2/dx^2 + \omega^2$ where ω is a constant and $(a, b) = (0, 1)$. We want to solve

$$\begin{cases} \frac{d^2 G}{dx^2} + \omega^2 G = \delta(x - \xi) & 0 < x < 1 \\ G(0, \xi) = G(1, \xi) = 0. \end{cases}$$

We can easily solve the ODE on either side of $x = \xi$ and we get

$$G(x, \xi) = \begin{cases} A(\xi) \sin(\omega x) + B(\xi) \cos(\omega x) & 0 < x < \xi \\ C(\xi) \sin[\omega(x - 1)] + D(\xi) \cos[\omega(x - 1)] & \xi < x < 1. \end{cases}$$

By the boundary conditions, $B(\xi) = D(\xi) = 0$. From continuity at $x = \xi$,

$$A(\xi) = C(\xi) \frac{\sin[\omega(\xi - 1)]}{\sin(\omega \xi)}.$$

From the jump discontinuity of dG/dx at $x = \xi$,

$$\begin{aligned} 1 &= \frac{C(\xi)\omega}{\sin(\omega \xi)} [\cos[\omega(\xi - 1)] \sin(\omega \xi) - \cos(\omega \xi) \sin[\omega(\xi - 1)]] \\ &= \frac{C(\xi)\omega}{\sin(\omega \xi)} \sin[\omega \xi - \omega(\xi - 1)] \\ &= \frac{C(\xi)\omega}{\sin(\omega \xi)} \sin(\omega). \end{aligned}$$

That is,

$$C(\xi) = \frac{\sin(\omega \xi)}{\omega \sin(\omega)} \text{ and } A(\xi) = \frac{\sin[\omega(\xi - 1)]}{\omega \sin(\omega)}.$$

Hence,

$$\begin{aligned} G(x, \xi) &= \begin{cases} \sin(\omega x) \sin[\omega(\xi - 1)] / (\omega \sin \omega) & 0 < x < \xi \\ \sin(\omega \xi) \sin[\omega(x - 1)] / (\omega \sin \omega) & \xi < x < 1 \end{cases} \\ &= \text{const.} \times \begin{cases} y_1(x) y_2(\xi) & 0 < x < \xi \\ y_1(\xi) y_2(x) & \xi < x < 1. \end{cases} \end{aligned}$$

Note that y_1 and y_2 are linearly independent solutions to $Ly = 0$ with $y_1(0) = y_2(1) = 0$. We see that the formula for G does not work when $\sin \omega = 0$. These values of ω correspond exactly with the eigenvalues of L . This is akin to solving systems of equations in matrix form, when we have $Ly = f$, the solution is $y = L^{-1}f$ when L is invertible. G is like L^{-1} , and when we are at the eigenvalues, $Ly = \lambda y$ so the kernel is non-trivial, and G is not defined.

4.3 A general result

Proposition 4.1

Let y_1 and y_2 be linearly independent solutions to $Ly = 0$ with $y_1(a) = y_2(b) = 0$. Then,

$$G(x, \xi) = \frac{1}{\alpha(\xi)(W(y_1, y_2))(\xi)} \times \begin{cases} y_1(x)y_2(\xi) & a < x < \xi \\ y_1(\xi)y_2(x) & \xi < x < b \end{cases}$$

satisfies $L[G(x, \xi)] = \delta(x - \xi)$ and $G(a, \xi) = G(b, \xi) = 0$.

Proof. We have $L[G(x, \xi)] = 0$ for $x \neq \xi$. Clearly, $G(\xi^+, \xi) = G(\xi^-, \xi)$. Also,

$$\begin{aligned} \left[\frac{dG}{dx} \right]_{\xi^+} - \left[\frac{dG}{dx} \right]_{\xi^-} &= \frac{1}{\alpha(\xi)(W(y_1, y_2))(\xi)} \underbrace{[y_1(\xi)y_2'(\xi) - y_1'(\xi)y_2(\xi)]}_{= (W(y_1, y_2))(\xi)} \\ &= \frac{1}{\alpha(\xi)}. \end{aligned}$$

Hence, $G(x, \xi)$ has all the desired properties for the Dirichlet Green's function. $\square$

So, the solution to the original problem is

$$\begin{aligned} y(x) &= \int_a^b G(x, \xi) f(\xi) d\xi \\ &= y_2(x) \int_a^x \frac{y_1(\xi)f(\xi)}{\alpha(\xi)(W(y_1, y_2))(\xi)} d\xi + y_1(x) \int_x^b \frac{y_2(\xi)f(\xi)}{\alpha(\xi)(W(y_1, y_2))(\xi)} d\xi. \end{aligned}$$

Example 4.2

Let us redo the last example. Take $L = d^2/dx^2 + \omega^2$ and $(a, b) = (0, 1)$. Use $y_1 = \sin(\omega x)$ and $y_2(x) = \sin[\omega(x - 1)]$. $\alpha(\xi) = 1$ and

$$\begin{aligned} (W(y_1, y_2))(\xi) &= \omega \sin(\omega \xi) \cos[\omega(\xi - 1)] - \omega \cos(\omega \xi) \sin[\omega(\xi - 1)] \\ &= \omega \sin[\omega \xi - \omega(\xi - 1)] \\ &= \omega \sin \omega. \end{aligned}$$

Hence, we get the same answer as before.

4.4 Green's functions for Sturm-Liouville operators

Suppose L has the form

$$L = \frac{1}{w} \left[-\frac{d}{dx} \left(p(x) \frac{d}{dx} \right) + q(x) \right].$$

Since we are interested in solutions to $Ly = f$, we can take $w = 1$ without loss of generality by changing f to wf . Note that for Sturm-Liouville operators, $\alpha(x) = -p(x)$.

Proposition 4.2

If L is a Sturm-Liouville operator and y_1, y_2 satisfies $Ly_1 = Ly_2 = 0$, then

$$p(x)(W(y_1, y_2)(x)) = \text{const.}$$

Proof. We have

$$\begin{aligned} y_2 Ly_1 - y_1 Ly_2 &= y_2(-(py_1')' + qy_1) - y_1(-(py_2')' + qy_2) \\ &= y_1(py_2')' - y_2(py_1')' \\ &= [p(y_1 y_2' - y_1' y_2)]'. \end{aligned}$$

So, if $Ly_1 = Ly_2 = 0$, we get $p(y_1 y_2' - y_1' y_2) = \text{const.}$ □

So, if L is a Sturm-Liouville operator,

$$G(x, \xi) = \text{const.} \times \begin{cases} y_1(x)y_2(\xi) & a < x < \xi \\ y_1(\xi)y_2(x) & \xi < x < b. \end{cases}$$

Then, the solution to $Ly = f$ with $y(a) = y(b) = 0$ is

$$y(x) = -\frac{1}{p(c)(W(y_1, y_2))(c)} \left[y_2(x) \int_a^x y_1(\xi) f(\xi) d\xi + y_1(x) \int_x^b y_2(\xi) f(\xi) d\xi \right]$$

where $c \in (a, b)$.

4.5 Eigenfunction expansions revisited

Recall that if L is a Sturm-Liouville operator on $V = \{y \in C^2([a, b]) : y(a) = y(b) = 0\}$, then there exist normalised eigenfunctions $\{Y_k\}_{k=1}^\infty$ such that $LY_k = \lambda_k Y_k$ and $\langle Y_k, Y_l \rangle = \delta_{kl}$. For any $f \in V$, we could write

$$f(x) = \sum_{k=1}^\infty \hat{f}_k Y_k(x) \text{ where } \hat{f}_k = \langle f, Y_k \rangle = \int_a^b f(x) Y_k(x) dx.$$

Note that $G(a, \xi) = G(b, \xi) = 0$, but $G(x, \xi) \notin V$ since it has a jump discontinuity. Instead, consider the sequence $\{G_n(x, \xi)\}_{n=1}^\infty$ such that $L[G_n(x, \xi)] = \delta_n(x - \xi)$ with $G_n(a, \xi) = G_n(b, \xi) = 0$. Each $G_n(x, \xi) \in V$ for any $x \in (a, b)$ so we can expand each $G_n(x, \xi)$. Taking the limit, we get

$$G(x, \xi) = \sum_{k=1}^\infty \hat{G}_k(\xi) Y_k(x)$$

and we pretend that $G(x, \xi) \in V$.

Proposition 4.3

If $\{Y_k\}$ are as above, then the Dirichlet Green's functions for the Sturm-Liouville operator L satisfies

$$G(x, \xi) = \sum_{k=1}^\infty \frac{Y_k(x) Y_k(\xi)}{\lambda_k}$$

where $LY_k = \lambda_k Y_k$.

Proof. By completeness, $G(x, \xi) = \sum_{k=1}^{\infty} \hat{G}_k(\xi) Y_k(x)$ where

$$\begin{aligned} \hat{G}_k(\xi) &= \langle G(\cdot, \xi), Y_k \rangle \\ &= \frac{1}{\lambda_k} \langle G(\cdot, \xi), LY_k \rangle \\ &= \frac{1}{\lambda_k} \langle LG(\cdot, \xi), Y_k \rangle \text{ since } L \text{ is self-adjoint} \\ &= \frac{1}{\lambda_k} \langle \delta(\cdot - \xi), Y_k \rangle \\ &= \frac{1}{\lambda_k} \int_a^b \delta(x - \xi) Y_k(x) dx \\ &= \frac{Y_k(\xi)}{\lambda_k} \end{aligned}$$

and we get the result. $\square$

If we have $\lambda_k = 0$ for some k , then $0 \neq \tilde{y} \in V$ with $L\tilde{y} = 0$. If $Ly = f$, then $L(y + \tilde{y}) = f$ as well, and there is no unique solution for y .

Example 4.3

Take $L = -d^2/dx^2$ and $(a, b) = (0, 1)$. We have $V = \{y \in C^2([0, 1]) : y(0) = y(1) = 0\}$. The normalised eigenfunctions are $Y_k(x) = \sqrt{2} \sin(k\pi x)$ with $\lambda_k = k^2\pi^2$ for $k \in \mathbb{Z}_{\geq 1}$. So,

$$G(x, \xi) = \sum_{k=1}^{\infty} \frac{2 \sin(k\pi x) \sin(k\pi \xi)}{k^2 \pi^2}.$$

We can check that this satisfies all the properties of Green's functions.

Applying L to the eigenfunction expansion for $G(x, \xi)$, we have

$$\delta(x - \xi) = L[G(x, \xi)] = \sum_{k=1}^{\infty} \frac{Y_k(\xi) L[Y_k(x)]}{\lambda_k} = \sum_{k=1}^{\infty} Y_k(\xi) Y_k(x)$$

which is the same formula in Section 4.1.2.

5 The Fourier transform

5.1 Definitions and simple properties

The Fourier transform is the $L \rightarrow \infty$ limit of the theory of Fourier series.

Definition 5.1 – Fourier transform

For $f : \mathbb{R} \rightarrow \mathbb{C}$, define its Fourier transform by

$$\hat{f}(\lambda) = \int_{-\infty}^{\infty} e^{-i\lambda x} f(x) dx$$

where $\lambda \in \mathbb{R}$.

We will use $\mathcal{F}$ to denote the relevant linear map, that is $\hat{f} = \mathcal{F}[f]$. We will occasionally write $\mathcal{F}_{x \rightarrow \lambda}[f(x)] = \hat{f}(\lambda)$. We will simplify and write $\int = \int_{-\infty}^{\infty}$. The Riemann-Lebesgue lemma also holds, that is if g is integrable

on $\mathbb{R}$, we have

$$\int e^{-i\lambda x} g(x) dx \rightarrow 0$$

as $|\lambda| \rightarrow \infty$.

Proposition 5.1

(i) Differentiation and multiplication

$$\mathcal{F}_{x \rightarrow \lambda} \left[\left(\frac{d}{dx} \right)^k f(x) \right] = (i\lambda)^k \hat{f}(\lambda) \text{ and } \mathcal{F}_{x \rightarrow \lambda} [x^k f(x)] = \left(i \frac{d}{d\lambda} \right)^k \hat{f}(\lambda).$$

(ii) Translation and phase shifts

$$\mathcal{F}_{x \rightarrow \lambda} [f(x - a)] = e^{-i\lambda a} \hat{f}(\lambda) \text{ and } \mathcal{F}_{x \rightarrow \lambda} [e^{-iax} f(x)] = \hat{f}(\lambda + a).$$

Proof. For (i),

$$\begin{aligned} \int f^{(k)}(x) e^{-i\lambda x} dx &= \int f(x) \left(-\frac{d}{dx} \right)^k e^{-i\lambda x} dx \text{ by IBP} \\ &= (i\lambda)^k \hat{f}(\lambda) \end{aligned}$$

and

$$\begin{aligned} \left(i \frac{d}{d\lambda} \right)^k \int e^{-i\lambda x} f(x) dx &= \int e^{-i\lambda x} x^k f(x) dx \\ &= \mathcal{F}_{x \rightarrow \lambda} [x^k f(x)]. \end{aligned}$$

The other two are left as exercises. □

Property (i) is important in solving ODEs. Consider

$$\left[P \left(\frac{d}{dx} \right) \right] y = F(x)$$

with $x \in \mathbb{R}$ where P is a polynomial (if $P(z) = z^2 + 1$, then $P(d/dx) = d^2/dx^2 + 1$). Taking the Fourier transform,

$$P(i\lambda) \hat{y}(\lambda) = \hat{F}(\lambda) \implies \hat{y}(\lambda) = \frac{\hat{F}(\lambda)}{P(i\lambda)}.$$

Can we get y back from $\hat{y}$?

Proposition 5.2

We can reconstruct f from $\hat{f}$ via the inversion formula

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\lambda x} \hat{f}(\lambda) d\lambda.$$

Proof. Write $\int = \lim_{n \rightarrow \infty} \int_{-n}^n$. The integral is the limit of

$$\begin{aligned} \frac{1}{2\pi} \int_{-n}^n e^{i\lambda x} \left[\int e^{-i\lambda y} f(y) dy \right] d\lambda &= \int f(y) \left[\frac{1}{2\pi} \int_{-n}^n e^{i\lambda(x-y)} d\lambda \right] dy \\ &= \int f(y) \frac{\sin[n(x-y)]}{\pi(x-y)} dy \\ &= \int f(x+y) \frac{\sin(ny)}{\pi y} dy. \end{aligned}$$

Recall from some example sheet last year, for all $n > 0$,

$$\int \frac{\sin(ny)}{\pi y} dy = 1.$$

So

$$\begin{aligned} \frac{1}{2\pi} \int_{-n}^n e^{i\lambda x} \hat{f}(\lambda) d\lambda - f(x) &= \int \sin(ny) \underbrace{\left[\frac{f(x+y) - f(x)}{\pi y} \right]}_{\text{call this } F(y;x)} dy \\ &= \frac{1}{n} \int \cos(ny) F_y(y;x) dy \rightarrow 0 \end{aligned}$$

□

Removing the limits in the first line of the proof,

$$f(x) = \int f(y) \left[\frac{1}{2\pi} \int e^{i\lambda(x-y)} d\lambda \right] dy,$$

that is

$$\delta(x-y) = \frac{1}{2\pi} \int e^{i\lambda(x-y)} d\lambda.$$

Proposition 5.3

For $f, g : \mathbb{R} \rightarrow \mathbb{C}$,

$$\int f(x) \overline{g(x)} dx = \frac{1}{2\pi} \int \hat{f}(\lambda) \overline{\hat{g}(\lambda)} d\lambda.$$

Hence,

$$\int |f(x)|^2 dx = \frac{1}{2\pi} \int |\hat{f}(\lambda)|^2 d\lambda.$$

Proof.

$$\begin{aligned} \frac{1}{2\pi} \int \hat{f}(\lambda) \overline{\hat{g}(\lambda)} d\lambda &= \frac{1}{2\pi} \int \left[\int e^{-i\lambda y} f(y) dy \right] \left[\int e^{i\lambda x} \overline{g(x)} dx \right] d\lambda \\ &= \int \int f(y) \overline{g(x)} \underbrace{\left[\frac{1}{2\pi} \int e^{i\lambda(x-y)} d\lambda \right]}_{= \delta(x-y)} dx dy \\ &= \int f(x) \overline{g(x)} dx \end{aligned}$$

□

Definition 5.2 – Convolution

For $f, g : \mathbb{R} \rightarrow \mathbb{C}$, define their convolution $f * g : \mathbb{R} \rightarrow \mathbb{C}$ by

$$(f * g)(x) = \int f(x - y)g(y) dy.$$

Note that $f * g = g * f$.

Proposition 5.4

$$\mathcal{F}_{x \rightarrow \lambda}[(f * g)(x)] = \hat{f}(\lambda)\hat{g}(\lambda)$$

Proof.

$$\begin{aligned} \mathcal{F}_{x \rightarrow \lambda}[(f * g)(x)] &= \int e^{-i\lambda x} \left[\int f(x - y)g(y) dy \right] dx \\ &= \int \int e^{-i\lambda x} f(x - y)g(y) dx dy \\ &= \int \int e^{-i\lambda(X+Y)} f(X)g(Y) dX dY \text{ where } X = x - y \text{ and } Y = y \\ &= \hat{f}(\lambda)\hat{g}(\lambda) \end{aligned}$$

□

5.2 Important examples**Example 5.1 – Exponentials**

Let $\sigma \in \mathbb{C}$ with $\operatorname{Re}(\sigma) > 0$. Let

$$f(x) = H(x)e^{-\sigma x} = \begin{cases} e^{-\sigma x} & x \geq 0 \\ 0 & x < 0. \end{cases}$$

Then,

$$\hat{f}(\lambda) = \int f(x)e^{-i\lambda x} dx = \int_0^\infty e^{-(\sigma+i\lambda)x} dx = \frac{1}{\sigma+i\lambda}.$$

Note that $|\hat{f}(\lambda)| \rightarrow 0$ as $\lambda \rightarrow \infty$. This does not decay very quickly since the original function is not very smooth. From the inversion formula,

$$H(x)e^{-\sigma x} = \frac{1}{2\pi} \int \frac{e^{i\lambda x}}{\sigma+i\lambda} d\lambda.$$

Differentiating with respect to σ , we have

$$H(x)x^k e^{-\sigma x} = \left(-\frac{\partial}{\partial \sigma}\right)^k \frac{1}{2\pi} \int \frac{e^{i\lambda x}}{\sigma+i\lambda} d\lambda = \frac{k!}{2\pi} \int \frac{e^{i\lambda x}}{(\sigma+i\lambda)^{k+1}} d\lambda.$$

Example 5.2 – Gaussians

Consider

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

which is the probability density function for $\mathcal{N}(0, 1)$.

$$\hat{f}(\lambda) = \frac{1}{\sqrt{2\pi}} \int e^{-i\lambda x - x^2/2} dx.$$

This is tough without complex analysis or methods. Notice that $df/dx = -xf$ so

$$(i\lambda)\hat{f} = -\left(i\frac{d}{d\lambda}\right)\hat{f} \implies \frac{d\hat{f}}{d\lambda} = -\lambda\hat{f}.$$

Solving, we get $\hat{f}(\lambda) = \hat{f}(0)e^{-\lambda^2/2}$. From the definition, $\hat{f}(0) = \int f(x) dx = 1$. Hence, $\hat{f}(\lambda) = e^{-\lambda^2/2}$. So, Gaussians are eigenfunctions of the operator $\mathcal{F}$ since $\mathcal{F}[f] = \sqrt{2\pi}f$. Also, we have actually computed the tough integral.

Example 5.3 – Dirac delta function

From the definition of $\delta(x)$, we have

$$\hat{\delta}(\lambda) = \int e^{-i\lambda x} \delta(x) dx = 1.$$

From the inversion formula,

$$\delta(x) = \frac{1}{2\pi} \int e^{i\lambda x} d\lambda.$$

5.3 Initial value problems revisited

Recall from Section 4.6 the initial value problem

$$\begin{cases} Ly = f(t) & t > 0 \\ y(0) = \dot{y}(0) = 0. \end{cases}$$

Now, assume that L has constant coefficients so $L = \alpha d^2/dt^2 + \beta d/dt + \gamma$. Take $\alpha = 1$ without loss of generality. Extend the definition of $t \mapsto y(t)$ to all of $\mathbb{R}$ by setting $y(t) = 0$ when $t < 0$. The same for f . It is customary to write $\mathcal{F}_{t \rightarrow \omega}[y(t)] = \hat{y}(\omega)$ when dealing with time-like variables. Taking the Fourier transform,

$$[(i\omega)^2 + \beta(i\omega) + \gamma]\hat{y}(\omega) = \hat{f}(\omega) \implies \hat{y}(\omega) = \frac{\hat{f}(\omega)}{P(\omega)} \text{ where } P(\omega) = (i\omega)^2 + \beta(i\omega) + \gamma.$$

Write $P(\omega) = (i\omega + \sigma_1)(i\omega + \sigma_2)$ and assume β, γ are such that $\text{Re}(\sigma_1), \text{Re}(\sigma_2) > 0$.

Initially, assume $\sigma_1 \neq \sigma_2$. Note

$$\frac{1}{P(\omega)} = \frac{1}{\sigma_2 - \sigma_1} \left[\frac{1}{i\omega + \sigma_1} - \frac{1}{i\omega + \sigma_2} \right]$$

and by Example 5.1,

$$\frac{1}{P(\omega)} = \frac{1}{\sigma_2 - \sigma_1} [\mathcal{F}_{t \rightarrow \omega}[H(t)e^{-\sigma_1 t}] - \mathcal{F}_{t \rightarrow \omega}[H(t)e^{-\sigma_2 t}]].$$

If we define

$$R(t) = H(t) \left[\frac{e^{-\sigma_1 t} - e^{-\sigma_2 t}}{\sigma_2 - \sigma_1} \right]$$

then $\hat{R}(\omega) = 1/P(\omega)$ so $\hat{y}(\omega) = \hat{R}(\omega)\hat{f}(\omega)$. We call $R(t)$ the response function for L . If $\sigma_1 = \sigma_2$, set $\sigma_1 = \sigma$ and $\sigma_2 = \sigma + \varepsilon$. Taking the limit $\varepsilon \rightarrow 0$, we get $R(t) = H(t)te^{-\sigma t}$.

In either case, by the convolution theorem,

$$y(t) = (R * f)(t) = \int R(t - \tau)f(\tau) d\tau = \int_0^t R(t - \tau)f(\tau) d\tau.$$

That is, $R(t - \tau) = G(t, \tau)$. What about the initial conditions? We did

$$\int_0^\infty (\ddot{y} + \beta\dot{y} + \gamma)y e^{-i\omega t} dt = \int_0^\infty e^{-i\omega t} f dt.$$

We then turned the differentiation into multiplication, where we did integration by parts in the background, and there we have applied the initial conditions.

Example 5.4

Consider a damped harmonic oscillator

$$\begin{cases} \ddot{y} + 2\gamma\dot{y} + y = f(t) & t > 0 \\ y(0) = \dot{y}(0) = 0 \end{cases}$$

where $\gamma > 0$. By the Fourier transform,

$$\hat{y}(\omega) = \frac{\hat{f}(\omega)}{P(\omega)} \text{ where } P(\omega) = \left[i\omega + \gamma + \sqrt{\gamma^2 - 1} \right] \left[i\omega + \gamma - \sqrt{\gamma^2 - 1} \right].$$

Note that $\text{Re}[\sigma_\pm(\gamma)] = \text{Re}[\gamma \pm \sqrt{\gamma^2 - 1}] > 0$.

So, in the case of $\gamma > 1$,

$$R(t) = H(t) \left[\frac{e^{-\sigma_- t} - e^{-\sigma_+ t}}{\sigma_+ - \sigma_-} \right] = H(t) e^{-\gamma t} \frac{\sinh[t\sqrt{\gamma^2 - 1}]}{\sqrt{\gamma^2 - 1}}.$$

This is overdamping. In the case $0 < \gamma < 1$,

$$R(t) = H(t) e^{-\gamma t} \frac{\sin[t\sqrt{1 - \gamma^2}]}{\sqrt{1 - \gamma^2}}.$$

This is underdamping. When $\gamma = 1$, take $\gamma = 1 + \varepsilon$ and send $\varepsilon \rightarrow 0$, we get $R(t) = H(t)te^{-t}$. This is critical damping.

5.4 Some neat examples

5.4.1 Poisson summation formula

Proposition 5.5 – Poisson summation formula

$$\sum_n f(x + n) = \sum_n \hat{f}(2\pi n) e^{2\pi i n x}$$

Proof. Notice both sides are periodic with period 1 so they have a Fourier series. Say the left hand side is

$F(x)$ and the right is $G(x)$, we have $\hat{G}_n = \hat{f}(2\pi n)$. F has Fourier coefficients

$$\begin{aligned}\hat{F}_n &= \int_0^1 e^{-2\pi i n x} F(x) dx \\ &= \sum_m \int_0^1 e^{-2\pi i n x} f(x+m) dx \\ &= \sum_m \int_m^{m+1} e^{-2\pi i n x} f(x) dx \\ &= \int_{-\infty}^{\infty} e^{-2\pi i n x} f(x) dx \\ &= \hat{f}(2\pi n)\end{aligned}$$

so $\hat{F}_n = \hat{G}_n$. That is, $F(x) = G(x)$. □

Example 5.5

Try $f(x) = H(x)xe^{-\sigma x}$ with $\text{Re}(\sigma) > 0$. Then, $\hat{f}(\lambda) = (\sigma + i\lambda)^{-2}$. Using the Poisson summation formula with $x = 0$, we have

$$\begin{aligned}\sum_n \hat{f}(2\pi n) &= \sum_n f(n) \\ \sum_n \frac{1}{(2\pi i n + \sigma)^2} &= \sum_{n=1}^{\infty} n e^{-\sigma n} = \frac{e^{\sigma}}{(1 - e^{\sigma})^2} = \frac{4}{\sinh^2(\sigma/2)}.\end{aligned}$$

Example 5.6

Consider

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

so $\hat{f}(\lambda) = e^{-\lambda^2/2}$. Hence,

$$\frac{1}{\sqrt{2\pi}} \sum_n e^{-(x+n)^2/2} = \sum_n e^{-(2\pi n)^2/2} e^{2\pi i n x}.$$

5.4.2 Heisenberg's uncertainty principle

Proposition 5.6

If $\int |\psi(x)|^2 dx = 1$, then

$$\int x^2 |\psi(x)|^2 dx \int \lambda^2 |\hat{\psi}(\lambda)|^2 d\lambda \geq \frac{\pi}{2}.$$

Proof. Integrating by parts,

$$1 = - \int x \frac{d}{dx} [\psi \bar{\psi}] dx = - \int (x \psi \bar{\psi}' + x \psi' \bar{\psi}) dx.$$

Then, by the triangle inequality,

$$1 \leq 2 \int x |\psi| |\psi'| dx \leq 2 \left(\int x^2 |\psi|^2 dx \right)^{1/2} \left(\int |\psi'|^2 dx \right)^{1/2}$$

using Cauchy-Schwarz. Recall that $\mathcal{F}_{x \rightarrow \lambda}[f'(x)] = (i\lambda)\hat{f}(\lambda)$ and $\int |f(x)|^2 dx = (1/2\pi) \int |\hat{f}(\lambda)|^2 d\lambda$. So, we

get

$$\int |f'(x)|^2 dx = \frac{1}{2\pi} \int \lambda^2 |\hat{f}(\lambda)|^2 d\lambda.$$

Combining them, we arrive at the result. □

We can replace $\psi(x)$ with $e^{-i\lambda_0 x} \psi(x + x_0)$, then $\hat{\psi}(\lambda) = e^{i\lambda x_0} \hat{\psi}(\lambda + \lambda_0)$, so

$$\int x^2 |\psi(x + x_0)|^2 dx \int \lambda^2 |\hat{\psi}(\lambda + \lambda_0)|^2 d\lambda = \int (x - x_0)^2 |\psi(x)|^2 dx \int (\lambda - \lambda_0)^2 |\hat{\psi}(\lambda)|^2 d\lambda \geq \frac{\pi}{2}.$$

In quantum mechanics,

$$\mathbb{P}(\text{particle is in } [a, b]) = \int_a^b |\psi(x)|^2 dx.$$

Also, up to a constant,

$$\mathbb{P}(\text{particle's momentum is in } [A, B]) = \int_A^B |\hat{\psi}(\lambda)|^2 d\lambda.$$

If we choose $x_0 = \int x |\psi(x)|^2 dx$ and $\lambda_0 = \int \lambda |\hat{\psi}(\lambda)|^2 d\lambda$, then $\Delta x \Delta p \geq \text{const.}$ where Δ is the variance or uncertainty.

5.4.3 From Fourier series to Fourier transforms

Consider smooth $f : \mathbb{R} \rightarrow \mathbb{C}$ and assume $f(x) = 0$ outside $|x| \leq L$. On the interval $(-L, L)$, we have

$$f(x) = \sum_n \hat{f}_n e^{i\pi n x / L} \text{ where } \hat{f}_n = \frac{1}{2L} \int_{-L}^L e^{-i\pi n x / L} f(x) dx = \frac{1}{2L} \int_{-\infty}^{\infty} e^{-i\pi n x / L} f(x) dx.$$

Write $\lambda_n = n\pi/L$ and $\delta\lambda_n = \lambda_{n+1} - \lambda_n = \pi/L$. Then, $\hat{f}_n = (1/2\pi) \delta\lambda_n \hat{f}(\lambda_n)$. So, for $x \in (-L, L)$,

$$f(x) = \frac{1}{2\pi} \sum_n e^{i\lambda_n x} \hat{f}(\lambda_n) \delta\lambda_n.$$

Taking $L \rightarrow \infty$, that is $\delta\lambda_n \rightarrow 0$, we get

$$f(x) = \frac{1}{2\pi} \int e^{i\lambda x} \hat{f}(\lambda) d\lambda$$

which is just the Fourier inversion formula. Also, from Parseval's theorem

$$\int_{-L}^L |f(x)|^2 dx = 2L \sum_n |\hat{f}_n|^2 = \frac{1}{2\pi} \sum_n |\hat{f}(\lambda_n)|^2 \delta\lambda_n,$$

so as $L \rightarrow \infty$, we get

$$\int |f(x)|^2 dx = \frac{1}{2\pi} \int |\hat{f}(\lambda)|^2 d\lambda.$$

That is, Parseval for Fourier series gives Parseval for Fourier transform.

5.4.4 Central limit theorem

Let $X_1, \dots, X_n$ be independent and identically distributed random variables with $\mathbb{E}[X_1] = \mu = 0$ and $\text{Var}(X_1) = \sigma^2 = 1$. Define $S_n = (X_1 + \dots + X_n)/\sqrt{n}$. Denote the probability density function of S_n by $f_n(x)$. The charac-

teristic function of S_n is

$$\begin{aligned}\hat{f}_n(\lambda) &= \mathbb{E}[e^{-i\lambda S_n}] \\ &= \mathbb{E}[e^{-i\lambda(X_1 + \dots + X_n)/\sqrt{n}}] \\ &= \mathbb{E}[e^{-i\lambda X_1/\sqrt{n}}] \dots \mathbb{E}[e^{-i\lambda X_n/\sqrt{n}}] \\ &= \mathbb{E}[e^{-i\lambda X_1/\sqrt{n}}]^n \\ &= \varphi_{X_1}(\lambda/\sqrt{n})^n\end{aligned}$$

where φ_{X_1} is the characteristic function of X_1 . Notice that $\varphi_{X_1}(0) = 1$, $\varphi'_{X_1}(0) = -i\mu = 0$ and $\varphi''_{X_1}(0) = -\sigma^2 = -1$. So,

$$\hat{f}_n(\lambda) = \left(1 - \frac{1}{2} \frac{\lambda^2}{n} + o(1/n)\right)^n = e^{-\lambda^2/2} + o(1) \text{ as } n \rightarrow \infty.$$

By Example 5.2, this is exactly the Fourier transform of the Gaussian. By the inverse Fourier transform,

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{1}{2\pi} \int e^{i\lambda x} \hat{f}_n(\lambda) d\lambda = \frac{1}{\sqrt{2\pi}} e^{-x^2/2},$$

that is, $(X_1 + \dots + X_n)/\sqrt{n} \rightarrow \mathcal{N}(0, 1)$ in distribution.

5.5 Discrete Fourier transform

Definition 5.3 – Discrete Fourier transform

Given a sequence $\{X_n\}_{n=0}^{N-1}$, we can define a new sequence $\{\hat{X}_k\}_{k=0}^{N-1}$ by the discrete Fourier transform

$$\hat{X}_k = \sum_{n=0}^{N-1} X_n e^{-i(2\pi n/N)k}.$$

Let $f : [0, 2\pi) \rightarrow \mathbb{C}$ be given and suppose $X_n = f(x_n)$ where $x_n = 2\pi n/N \in [0, 2\pi)$. Set $\delta x_n = x_{n+1} - x_n = 2\pi/N$.

$$\hat{X}_k = \frac{N}{2\pi} \sum_{n=0}^{N-1} f(x_n) e^{-ikx_n} \delta x_n.$$

If we extend $f(x) = 0$ when $x \notin [0, 2\pi)$,

$$\hat{X}_k \simeq \frac{N}{2\pi} \int e^{-ikx} f(x) dx = \frac{N}{2\pi} \hat{f}(k).$$

That is, $\hat{f}(k) \simeq (2\pi/N) \hat{X}_k$ for large N .

Proposition 5.7

Given $\{\hat{X}_k\}_{k=0}^{N-1}$, we can recover $\{X_n\}_{n=0}^{N-1}$ via

$$X_n = \frac{1}{N} \sum_{k=0}^{N-1} \hat{X}_k e^{i(2\pi k/N)n}.$$

Proof. By direct computation,

$$\begin{aligned}\sum_{k=0}^{N-1} \hat{X}_k e^{i(2\pi n/N)k} &= \sum_{k=0}^{N-1} \sum_{m=0}^{N-1} X_m e^{2\pi i(n-m)k/N} \\ &= \sum_{m=0}^{N-1} X_m \sum_{k=0}^{N-1} \omega_N^{(n-m)k} \text{ where } \omega_N = e^{2\pi i/N}.\end{aligned}$$

Note that

$$\sum_{k=0}^{N-1} \omega_N^{(n-m)k} = \begin{cases} N & \text{if } n = m \\ \frac{1 - \omega_N^{(n-m)N}}{1 - \omega_N^{(n-m)}} = 0 & \text{if } n \neq m. \end{cases}$$

So,

$$\sum_{k=0}^{N-1} \hat{X}_k e^{i(2\pi n/N)k} = \sum_{m=0}^{N-1} X_m (N\delta_{nm}) = NX_n$$

which gives the result. $\square$

If you prefer vectors $\mathbf{X} \in \mathbb{C}^n$ with $\mathbf{X} = (X_0 \dots X_{N-1})^\top$, we have $\hat{\mathbf{X}} = (\hat{X}_0 \dots \hat{X}_{N-1})^\top = \mathcal{F}\mathbf{X}$ where $\mathcal{F}$ is a $n \times n$ matrix with $\mathcal{F}_{kn} = e^{-i(2\pi n/N)k}$. Also, $\mathcal{F}\mathcal{F}^\dagger = \mathcal{F}^\dagger\mathcal{F} = NI_N$. If we define $U = \mathcal{F}/\sqrt{n}$, then U is unitary, that is $UU^\dagger = U^\dagger U = I_N$ and $(U\mathbf{X}) \cdot (U\mathbf{Y}) = \mathbf{X} \cdot \mathbf{Y}$.

Example 5.7

Take $X_n = \binom{N-1}{n}$, then

$$\hat{X}_k = \sum_{n=0}^{N-1} \binom{N-1}{n} e^{-i(2\pi n/N)k} = (1 + e^{-2\pi i k/N})^{N-1}.$$

Proposition 5.8

$$\sum_{n=0}^{N-1} X_n \overline{Y_n} = \frac{1}{N} \sum_{k=0}^{N-1} \hat{X}_k \overline{\hat{Y}_k}$$

and in particular,

$$\sum_{n=0}^{N-1} |X_n|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |\hat{X}_k|^2.$$

Proof. Again, by direct computation,

$$\begin{aligned}\frac{1}{N} \sum_{k=0}^{N-1} \hat{X}_k \overline{\hat{Y}_k} &= \frac{1}{N} (\mathcal{F}\mathbf{X}) \cdot (\mathcal{F}\mathbf{Y}) \\ &= (U\mathbf{X}) \cdot (U\mathbf{Y}) \\ &= \mathbf{X} \cdot \mathbf{Y} \\ &= \sum_{n=0}^{N-1} X_n \overline{Y_n}.\end{aligned}$$

$\square$

Example 5.8

Take $X_n = \binom{N-1}{n}$ as before, then

$$\begin{aligned} \sum_{n=0}^{N-1} \binom{N-1}{n}^2 &= \frac{1}{N} \sum_{k=0}^{N-1} (1 + e^{-2\pi i k/N})^{N-1} (1 + e^{2\pi i k/N})^{N-1} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} (e^{i\pi k/N} + e^{-i\pi k/N})^{2N-2} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} \left(2 \cos \left(\frac{\pi k}{N} \right) \right)^{2N-2}. \end{aligned}$$

Note that

$$\sum_{n=0}^{N-1} \binom{N-1}{n}^2 = \sum_{n=0}^{N-1} \binom{N-1}{n} \binom{N-1}{N-1-n} = \binom{2N-2}{N-1}.$$

Setting $N-1 = M$, we get

$$\begin{aligned} \frac{1}{2^{2M}} \binom{2M}{M} &= \frac{1}{M+1} \sum_{k=0}^M \cos^{2M} \left(\frac{\pi k}{M+1} \right) \\ &= \mathbb{P}(\text{getting exactly } M \text{ heads after tossing a fair coin } 2M \text{ times}). \end{aligned}$$

Just like with the Fourier transform of functions $f : \mathbb{R} \rightarrow \mathbb{C}$, we can define

Definition 5.4 – Convolution of sequences

The convolution of two sequences $\{X_n\}_{n=0}^{N-1}$ and $\{Y_n\}_{n=0}^{N-1}$ is

$$(X * Y)_n = \sum_{m=0}^{N-1} X_{(n-m) \bmod N} Y_m.$$

Proposition 5.9

$$[\mathcal{F}(X * Y)]_k = \hat{X}_k \hat{Y}_k$$

Proof. Dropping the mod N ,

$$\begin{aligned} [\mathcal{F}(X * Y)]_k &= \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} X_{n-m} Y_m e^{-i(2\pi n/N)k} \\ &= \sum_{m=0}^{N-1} Y_m e^{-i(2\pi m/N)k} \sum_{n=0}^{N-1} X_{n-m} e^{-i(2\pi(n-m)/N)k} \\ &= \sum_{m=0}^{N-1} Y_m e^{-i(2\pi m/N)k} \hat{X}_k \\ &= \hat{X}_k \hat{Y}_k. \end{aligned}$$

□

Just by looking at the formula for the discrete Fourier transform, we might think that the complexity for computing the transform is $\mathcal{O}(N^2)$. But, there is the fast Fourier transform, if $N = 2^M$, then the complexity drops to $\mathcal{O}(N \log_2 N)$. If we allow quantum computers, it drops further to $\mathcal{O}((\log_2 N)^2)$ by the quantum fast Fourier transform.

6 PDEs on unbounded domains

6.1 Well-posedness

Hadamard declared that a problem is well-posed if

- (i) a solution exists;
- (ii) the solution is unique;
- (iii) the solution depends continuously on the given data (initial conditions and boundary conditions).

(i) and (ii) are obvious, but (iii) is subtle. To mention continuity implicitly means we have certain norms/metrics/topologies in mind. For example, if we have an IBVP whose initial data and boundary data lie in X , and at time $t > 0$, the solution $u(\cdot, t)$ belongs to some Y . For each $t > 0$, we get the abstract map $S_t : X \rightarrow Y$. Continuity depends on d_X and d_Y . Intuitively, we want start close and stay close.

Example 6.1

Consider the IVP $dx/dt = -x$ with $x(0) = x_0$. Clearly, (i) and (ii) are satisfied. The solution is $X_0(t) = x_0 e^{-t}$. The solution with initial data $x(0) = x_1$ is $X_1(t) = x_1 e^{-t}$. So,

$$|X_1(t) - X_0(t)| = e^{-t} |x_1 - x_0| \leq |x_1 - x_0|.$$

In this case, the problem is well-posed.

Example 6.2

Consider the backwards heat equation IBVP on $\Omega = (0, \pi)$,

$$\begin{cases} \varphi_t + \varphi_{xx} = 0 & \Omega \times (0, \infty) \\ \varphi = 0 & \partial\Omega \times (0, \infty) \\ \varphi = f & \Omega \times \{t = 0\}. \end{cases}$$

If $f(x) = 0$, we get $\varphi(x, t) = 0$. If $f(x) = f_n(x) = \sin(nx)/n$. Then, $\varphi_n(x, t) = e^{n^2 t} \sin(nx)/n$. Then,

$$\|f - f_n\|_\infty = \sup_{(0, \pi)} |f(x) - f_n(x)| = \frac{1}{n} \rightarrow 0.$$

However,

$$\|\varphi(\cdot, t) - \varphi_n(\cdot, t)\|_\infty = \frac{1}{n} e^{n^2 t} \rightarrow \infty$$

so the problem is ill-posed (even locally in time).

6.2 The method of characteristics

We want to solve

$$a(x, y) \frac{\partial u}{\partial x} + b(x, y) \frac{\partial u}{\partial y} = c(x, y, u)$$

along with u satisfying some initial data on some curve C in $\mathbb{R}^2$. That is, we want $u(x, y) = \phi(x, y)$ when $(x, y) \in C$. Consider curves $(x, y) = (x(t), y(t))$ defined by

$$\frac{dx}{dt} = a(x, y) \text{ and } \frac{dy}{dt} = b(x, y)$$

with $(x(0), y(0)) \in C$. We call these the characteristic curves for the PDE. Solving them, we can get a whole family of solutions determined by the starting point $(x(0), y(0))$. Consider the evolution of $u(x, y)$ along a given characteristic, that is, set $z(t) = u(x(t), y(t))$. By chain rule,

$$\begin{aligned}\frac{dz}{dt} &= \frac{dx}{dt} \frac{\partial u}{\partial x} + \frac{dy}{dt} \frac{\partial u}{\partial y} \\ &= a(x, y) \frac{\partial u}{\partial x} + b(x, y) \frac{\partial u}{\partial y} \\ &= c(x, y, z).\end{aligned}$$

Note that $z(0) = u(x(0), y(0)) = \phi(x(0), y(0))$ since $(x(0), y(0)) \in C$. So,

$$\frac{dz}{dt} = c(x, y, z) \text{ with } z(0) = \phi(x(0), y(0)).$$

We need to invert the relationship between (x, y) and (s, t) where the curve C has been parameterised by s .

Example 6.3

Consider

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = u$$

with $u(x, 0) = f(x)$ so $C = \{(s, 0) : s \in \mathbb{R}\}$. The characteristic curves satisfy

$$\frac{dx}{dt} = 1 \text{ and } \frac{dy}{dt} = 1 \implies \begin{cases} x = t + x_0 \\ y = t + y_0. \end{cases}$$

We want $(x_0, y_0) \in C$ so we take $(x_0, y_0) = (s, 0)$. Then, the curves are $(x, y) = (t + s, t)$. Also,

$$\frac{dz}{dt} = z \implies z(t) = z_0 e^t$$

and $z_0 = z(0) = u(x_0, y_0) = u(s, 0) = f(s)$. Hence, $z(t, s) = f(s)e^t$. Inverting the relationship between (x, y) and (s, t) , we get $t = y$ and $s = x - y$. Combining, we get

$$u(x, y) = z(t(x, y), s(x, y)) = f(x - y)e^y.$$

Example 6.4

Consider

$$(1 + x^2) \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = u + 1$$

with $u(0, y) = f(y)$ so $C = \{(0, s) : s \in \mathbb{R}\}$. The characteristic curves

$$\frac{dx}{dt} = 1 + x^2 \text{ and } \frac{dy}{dt} = 1 \implies \begin{cases} x(t) = \tan(t + \arctan x_0) \\ y(t) = t + y_0. \end{cases}$$

We want $(x_0, y_0) \in C$, that is $(x_0, y_0) = (0, s)$. So, the curves are $(x, y) = (\tan t, t + s)$. Also,

$$\frac{dz}{dt} = z + 1 \implies z(t) = -1 + (z_0 + 1)e^t$$

and $z_0 = u(x_0, y_0) = u(0, s) = f(s)$. Hence, $z(t, s) = -1 + (f(s) + 1)e^t$. Inverting, we get $t = \arctan x$ and $s = y - \arctan x$. Combining, we get

$$u(x, y) = -1 + (f(y - \arctan x) + 1)e^{\arctan x}.$$

6.3 Classification of second order linear PDEs in two variables

Consider the generic problem $Lu = y$ where

$$Lu = a(x, y) \frac{\partial^2 u}{\partial x^2} + 2b(x, y) \frac{\partial^2 u}{\partial x \partial y} + c(x, y) \frac{\partial^2 u}{\partial y^2} + d(x, y) \frac{\partial u}{\partial x} + e(x, y) \frac{\partial u}{\partial y} + f(x, y)u.$$

We want to introduce $\xi = \xi(x, y)$ and $\eta = \eta(x, y)$ such that the PDE simplifies. Write $(x, y) = (x_1, x_2)$ and $(\xi, \eta) = (\xi_1, \xi_2)$. Then,

$$Lu = \sum_{i,j=1}^2 a_{ij}(x, y) \frac{\partial^2 u}{\partial x_i \partial x_j} + [\text{lower order stuff}]$$

where

$$(a_{ij}) = \begin{pmatrix} a(x, y) & b(x, y) \\ b(x, y) & c(x, y) \end{pmatrix}.$$

Proposition 6.1

If $(x, y) \mapsto (\xi, \eta)$ and $U(\xi, \eta) = u(x, y)$, then Lu becomes

$$\tilde{L}U = \sum_{p,q=1}^2 A_{pq} \frac{\partial^2 U}{\partial \xi_p \partial \xi_q} + [\text{lower order bits}]$$

where

$$A_{pq} = \sum_{i,j=1}^2 a_{ij} \frac{\partial \xi_p}{\partial x_i} \frac{\partial \xi_q}{\partial x_j}.$$

Proof. We adopt the summation convention and use the chain rule. $\partial u / \partial x_i = (\partial \xi_p / \partial x_i)(\partial U / \partial \xi_p)$ so

$$\frac{\partial^2 u}{\partial x_i \partial x_j} = \frac{\partial \xi_p}{\partial x_i} \frac{\partial \xi_q}{\partial x_j} \frac{\partial^2 U}{\partial \xi_p \partial \xi_q} + \frac{\partial^2 \xi_p}{\partial x_i \partial x_j} \frac{\partial U}{\partial \xi_p}$$

which gives

$$a_{ij} \frac{\partial^2 u}{\partial x_i \partial x_j} = A_{pq} \frac{\partial^2 U}{\partial \xi_p \partial \xi_q} + [\text{lower order}].$$

□

We can read off that

$$\begin{aligned} A_{11} &= a\xi_x^2 + 2b\xi_x\xi_y + c\xi_y^2 \\ A_{12} &= a\xi_x\eta_y + b(\xi_x\eta_y + \xi_y\eta_x) + c\xi_y\eta_x \\ A_{22} &= a\eta_x^2 + 2b\eta_x\eta_y + c\eta_y^2. \end{aligned}$$

We can make $A_{11} = 0$ and $A_{22} = 0$ if $M = \xi_x / \xi_y$ and $N = \eta_x / \eta_y$ satisfy $az^2 + 2bz + c = 0$ with roots

$$z = \frac{-b \pm \sqrt{b^2 - ac}}{a}.$$

If M and N are chosen to satisfy this quadratic, we call the curves $\xi(x, y) = \text{const.}$ and $\eta(x, y) = \text{const.}$ the characteristic curves of the PDE. On the characteristic curves,

$$\frac{d}{dx} [\xi(x, y(x))] = \xi_x + \frac{dy}{dx} \xi_y = 0$$

and $\eta_x + (dy/dx)\eta_y = 0$. That is,

$$\frac{dy}{dx} = -\frac{\xi_x}{\xi_y} = -M \text{ and } \frac{dy}{dx} = -\frac{\eta_x}{\eta_y} = -N.$$

So, on the characteristic curves,

$$\frac{dy}{dx} = -\left[\frac{-b \pm \sqrt{b^2 - ac}}{a} \right].$$

This leads to a classification.

Definition 6.1

- (i) If $b^2 - ac < 0$, we say that the PDE is elliptic and there is no characteristic curves.
- (ii) If $b^2 - ac = 0$, we say that the PDE is parabolic and there is one family of characteristic curves.
- (iii) If $b^2 - ac > 0$, we say that the PDE is hyperbolic and there are two families of characteristic curves.

Laplace's equation is elliptic, the heat equation is parabolic, and the wave equation is hyperbolic.

In hyperbolic regions, we can integrate dy/dx to get the coordinates $\xi = \xi(x, y)$ and $\eta = \eta(x, y)$ for which $A_{11} = 0$ and $A_{22} = 0$. The partial differential operator becomes

$$2A_{12} \frac{\partial^2 U}{\partial \xi \partial \eta} + [\text{lower order terms}].$$

We say that the PDE in this form is in canonical form.

Example 6.5

Consider the wave equation with wave speed $c = 1$.

$$\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0.$$

Then, $a = 1$, $b = 0$ and $c = -1$. So, $b^2 - ac = 1 > 0$ and the wave equation is globally hyperbolic. To get the characteristic curves, we use $dy/dx = \mp 1$ and so $x \pm y = \text{const.}$. That is, we can take $\xi = x + y$ and $\eta = x - y$. In these coordinates,

$$4 \frac{\partial^2 U}{\partial \xi \partial \eta} + [\text{lower order terms} = 0] = 0.$$

We get $U(\xi, \eta) = A(\xi) + B(\eta)$ so $u(x, y) = A(x + y) + B(x - y)$. We can solve the IVP

$$\begin{cases} u_{tt} - u_{xx} = 0 & \mathbb{R} \times (0, \infty) \\ u = f & \mathbb{R} \times \{t = 0\} \\ u_t = g & \mathbb{R} \times \{t = 0\}. \end{cases}$$

We have $A(x) + B(x) = f(x)$ and $A'(x) - B'(x) = g(x)$ and get

$$u(x, t) = \frac{1}{2}[f(x + t) + f(x - t)] + \frac{1}{2} \int_{x-t}^{x+t} g(s) ds.$$

Example 6.6

Consider

$$xy \frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0.$$

Then, $a = xy$, $b = 0$ and $c = -1$. So, $b^2 - ac = xy$ and the hyperbolic region is $\{xy > 0\} = \{x, y > 0\} \cup \{x, y < 0\}$. On the characteristic curves, $dy/dx = \mp 1/\sqrt{xy}$ and integrating up (on $x, y > 0$, the other region is similar), we get $y^{3/2}/3 \pm x^{1/2} = \text{const.}$. So,

$$\xi(x, y) = \frac{1}{3}y^{3/2} + x^{1/2} \text{ and } \eta(x, y) = \frac{1}{3}y^{3/2} - x^{1/2}.$$

The PDE becomes

$$-\frac{1}{2}y \frac{\partial^2 U}{\partial \xi \partial \eta} + [\text{lower order terms}] = 0.$$

Note that $2y^{3/2}/3 = \xi + \eta$ so $y = [3(\xi + \eta)/2]^{2/3}$ and we can write the PDE only in terms of ξ and η .

6.4 Fourier transform in higher dimensions

For $f : \mathbb{R}^n \rightarrow \mathbb{C}$, define the Fourier transform pair

$$\hat{f}(\boldsymbol{\lambda}) = \int e^{-i\boldsymbol{\lambda} \cdot \mathbf{x}} f(\mathbf{x}) d^n \mathbf{x} \text{ and } f(\mathbf{x}) = \frac{1}{(2\pi)^n} \int e^{i\boldsymbol{\lambda} \cdot \mathbf{x}} \hat{f}(\boldsymbol{\lambda}) d^n \boldsymbol{\lambda}$$

where $\mathbf{x}, \boldsymbol{\lambda} \in \mathbb{R}^n$ and $\int = \int_{\mathbb{R}^n}$. For $f, g : \mathbb{R}^n \rightarrow \mathbb{C}$, define the convolution as

$$(f * g)(\mathbf{x}) = \int f(\mathbf{x} - \mathbf{y}) g(\mathbf{y}) d^n \mathbf{y}.$$

We have the convolution theorem $\mathcal{F}_{\mathbf{x} \rightarrow \boldsymbol{\lambda}}[(f * g)(\mathbf{x})] = \hat{f}(\boldsymbol{\lambda}) \hat{g}(\boldsymbol{\lambda})$. Also,

$$\mathcal{F}_{\mathbf{x} \rightarrow \boldsymbol{\lambda}} \left[\left(\frac{\partial}{\partial x_1} \right)^{\alpha_1} \cdots \left(\frac{\partial}{\partial x_n} \right)^{\alpha_n} f(\mathbf{x}) \right] = (i\lambda_1)^{\alpha_1} \cdots (i\lambda_n)^{\alpha_n} \hat{f}(\boldsymbol{\lambda}).$$

Example 6.7

If $\Delta = (\partial/\partial x_1)^2 + \cdots + (\partial/\partial x_n)^2$, then

$$\mathcal{F}_{\mathbf{x} \rightarrow \boldsymbol{\lambda}}[\Delta f(\mathbf{x})] = (-\lambda_1^2 - \cdots - \lambda_n^2) \hat{f}(\boldsymbol{\lambda}) = -|\boldsymbol{\lambda}|^2 \hat{f}(\boldsymbol{\lambda}).$$

We have the higher dimensional version of the Dirac delta function $\delta(\mathbf{x})$, defined by the properties $\delta(\mathbf{x}) = 0$ when $\mathbf{x} \neq \mathbf{0}$ and

$$\int_{|\mathbf{x}| < \varepsilon} \delta(\mathbf{x}) d^n \mathbf{x} = 1$$

for all $\varepsilon > 0$. So,

$$\int \delta(\mathbf{x} - \mathbf{y}) f(\mathbf{y}) d^n \mathbf{y} = f(\mathbf{x}).$$

We get

$$\hat{\delta}(\boldsymbol{\lambda}) = \int e^{-i\boldsymbol{\lambda} \cdot \mathbf{x}} \delta(\mathbf{x}) d^n \mathbf{x} = 1.$$

By inversion,

$$\delta(\mathbf{x}) = \frac{1}{(2\pi)^n} \int e^{-i\boldsymbol{\lambda} \cdot \mathbf{x}} d^n \boldsymbol{\lambda}.$$

Example 6.8

Similar to the 1D case, if

$$g(\mathbf{x}) = \frac{1}{(2\pi)^{n/2}} e^{-|\mathbf{x}|^2/2},$$

then $\hat{g}(\boldsymbol{\lambda}) = e^{-|\boldsymbol{\lambda}|^2/2}$. To see this,

$$\begin{aligned}\hat{g}(\boldsymbol{\lambda}) &= \frac{1}{(2\pi)^{n/2}} \int e^{-i\boldsymbol{\lambda} \cdot \mathbf{x}} e^{-|\mathbf{x}|^2/2} d^n \mathbf{x} \\ &= \int dx_1 \cdots \int dx_n \left[\frac{1}{\sqrt{2\pi}} e^{-i\lambda_1 x_1 - x_1^2/2} \right] \cdots \left[\frac{1}{\sqrt{2\pi}} e^{-i\lambda_n x_n - x_n^2/2} \right] \\ &= e^{-\lambda_1^2/2} \cdots e^{-\lambda_n^2/2} \\ &= e^{-|\boldsymbol{\lambda}|^2/2}.\end{aligned}$$

6.5 Green's function for the heat equation

We want to solve

$$\begin{cases} u_t - \kappa \Delta u = F(\mathbf{x}, t) & (\mathbf{x}, t) \in \mathbb{R}^n \times (0, \infty) \\ u(\mathbf{x}, 0) = f(\mathbf{x}) & \mathbf{x} \in \mathbb{R}^n. \end{cases}$$

- (i) Zero forcing ($F = 0$) and arbitrary initial data.
- (ii) Zero initial data ($f = 0$) and arbitrary forcing.

We can split the problem into two as above, then use superposition to add the two solutions together. We will make use of the heat kernel

$$K_t(\mathbf{x}) = \frac{1}{(4\pi\kappa t)^{n/2}} \exp \left[-\frac{|\mathbf{x}|^2}{4\kappa t} \right]$$

where $t > 0$.

Proposition 6.2

$$\hat{K}_t(\boldsymbol{\lambda}) = \mathcal{F}_{\mathbf{x} \rightarrow \boldsymbol{\lambda}}[K_t(\mathbf{x})] = e^{-\kappa t |\boldsymbol{\lambda}|^2}$$

Proof.

$$\hat{K}_t(\boldsymbol{\lambda}) = \int e^{-i\boldsymbol{\lambda} \cdot \mathbf{x}} K_t(\mathbf{x}) d^n \mathbf{x} = \frac{1}{(4\pi\kappa t)^{n/2}} \int e^{-i\boldsymbol{\lambda} \cdot \mathbf{x}} e^{-|\mathbf{x}|^2/(4\kappa t)} d^n \mathbf{x}.$$

Using the substitution $\mathbf{x} = \sqrt{2\kappa t} \mathbf{x}'$, then $d^n \mathbf{x} = (2\kappa t)^{n/2} d^n \mathbf{x}'$, so

$$\begin{aligned}\hat{K}_t(\boldsymbol{\lambda}) &= \frac{(2\kappa t)^{n/2}}{(4\pi\kappa t)^{n/2}} \int e^{-i\boldsymbol{\lambda} \cdot [\sqrt{2\kappa t} \mathbf{x}']} e^{-|\mathbf{x}'|^2/2} d^n \mathbf{x}' \\ &= \frac{1}{(2\pi)^{n/2}} \int e^{-i(\sqrt{2\kappa t} \boldsymbol{\lambda}) \cdot \mathbf{x}'} e^{-|\mathbf{x}'|^2/2} d^n \mathbf{x}' \\ &= \hat{g}(\sqrt{2\kappa t} \boldsymbol{\lambda}) \\ &= e^{-\kappa t |\boldsymbol{\lambda}|^2}\end{aligned}$$

where $g(\mathbf{x})$ is the Gaussian in Example 6.8. □

We see that $\hat{K}_t(\boldsymbol{\lambda}) \rightarrow 1$ as $t \rightarrow 0$. So, it looks like $K_t(\mathbf{x}) \rightarrow \delta(\mathbf{x})$ as $t \rightarrow 0$. We will solve the problems (i) and (ii) using the Fourier transform.

Proposition 6.3

The solution to (i) is $u(\mathbf{x}, t) = (K_t * f)(\mathbf{x})$.

Proof. Taking the Fourier transform of $u_t - \kappa \Delta u = 0$ and $u(\mathbf{x}, 0) = f(\mathbf{x})$. We get

$$\frac{\partial \hat{u}}{\partial t} + \kappa |\boldsymbol{\lambda}|^2 \hat{u} = 0 \text{ and } \hat{u}(\boldsymbol{\lambda}, 0) = \hat{f}(\boldsymbol{\lambda}).$$

Solving this ODE, we have

$$\hat{u}(\boldsymbol{\lambda}, t) = \hat{u}(\boldsymbol{\lambda}, 0) e^{-\kappa |\boldsymbol{\lambda}|^2 t} = \hat{f}(\boldsymbol{\lambda}) \hat{K}_t(\boldsymbol{\lambda}).$$

By the convolution theorem, $u(\mathbf{x}, t) = (K_t * f)(\mathbf{x})$. □

Proposition 6.4

The solution to (ii) is

$$u(\mathbf{x}, t) = \int_0^t \left[\int_{\mathbb{R}^n} K_{t-s}(\mathbf{x} - \mathbf{y}) F(\mathbf{y}, s) d^n \mathbf{y} \right] ds.$$

Proof. Taking the Fourier transform,

$$\frac{\partial \hat{u}}{\partial t} + \kappa |\boldsymbol{\lambda}|^2 \hat{u} = \hat{F}(\boldsymbol{\lambda}, t) \text{ and } \hat{u}(\boldsymbol{\lambda}, 0) = 0.$$

Using the integrating factor $e^{\kappa |\boldsymbol{\lambda}|^2 t}$, we get

$$\frac{\partial}{\partial t} [\hat{u} e^{\kappa |\boldsymbol{\lambda}|^2 t}] = \hat{F}(\boldsymbol{\lambda}, t) e^{\kappa |\boldsymbol{\lambda}|^2 t}.$$

Integrating with respect to t , we have

$$\begin{aligned} \hat{u}(\boldsymbol{\lambda}, t) e^{\kappa |\boldsymbol{\lambda}|^2 t} &= \int_0^t \hat{F}(\boldsymbol{\lambda}, s) e^{\kappa |\boldsymbol{\lambda}|^2 s} ds \\ \hat{u}(\boldsymbol{\lambda}, t) &= \int_0^t \hat{F}(\boldsymbol{\lambda}, s) e^{\kappa |\boldsymbol{\lambda}|^2 (t-s)} ds = \int_0^t \hat{K}_{t-s}(\boldsymbol{\lambda}) \hat{F}(\boldsymbol{\lambda}, s) ds. \end{aligned}$$

By the convolution theorem,

$$u(\mathbf{x}, t) = \int_0^t \left[\int_{\mathbb{R}^n} K_{t-s}(\mathbf{x} - \mathbf{y}) F(\mathbf{y}, s) d^n \mathbf{y} \right] ds. \quad \square$$

Hence, the solution to the original problem is

$$u(\mathbf{x}, t) = (K_t * f)(\mathbf{x}) + \int_0^t \left[\int_{\mathbb{R}^n} K_{t-s}(\mathbf{x} - \mathbf{y}) F(\mathbf{y}, s) d^n \mathbf{y} \right] ds.$$

Focus on problem (ii). Define $G(\mathbf{x}, t; \mathbf{y}, s) = H(t-s) K_{t-s}(\mathbf{x} - \mathbf{y})$. If we set $F(\mathbf{x}, t) = 0$ when $t < 0$, then the solution to (ii) is

$$u(\mathbf{x}, t) = \int_{-\infty}^{\infty} \left[\int_{\mathbb{R}^n} G(\mathbf{x}, t; \mathbf{y}, s) F(\mathbf{y}, s) d^n \mathbf{y} \right] ds.$$

We see that $G(\mathbf{x}, t; \mathbf{y}, s)$ is the Green's function for the heat equation with zero initial data. We can show that

$$\frac{\partial G}{\partial t} - \kappa \Delta G = \delta(t-s) \delta(\mathbf{x} - \mathbf{y}),$$

or equivalently,

$$\frac{\partial \hat{G}}{\partial t} + \kappa |\boldsymbol{\lambda}|^2 \hat{G} = e^{-i\boldsymbol{\lambda} \cdot \mathbf{y}} \delta(t-s)$$

by $\hat{G}(\boldsymbol{\lambda}, t; \mathbf{y}, s) = H(t-s) e^{-i\boldsymbol{\lambda} \cdot \mathbf{y}} \hat{K}_{t-s}(\boldsymbol{\lambda})$.

6.6 Green's function for Laplace's equation

We want to find $G = G(\mathbf{x}, \mathbf{y})$ such that for each fixed $\mathbf{y} \in \mathbb{R}^n$, we have

$$\Delta G(\mathbf{x}, \mathbf{y}) = \delta(\mathbf{x} - \mathbf{y}) \text{ and } G(\mathbf{x}, \mathbf{y}) \rightarrow 0 \text{ as } |\mathbf{x}| \rightarrow \infty.$$

This is called the free space Green's function for Laplace's equation. We can think of $G(\mathbf{x}, \mathbf{y})$ as the electric potential at $\mathbf{x}$ due to a point charge at $\mathbf{y}$. By taking the Fourier transform,

$$\hat{G}(\boldsymbol{\lambda}, \mathbf{y}) = -\frac{e^{-i\boldsymbol{\lambda} \cdot \mathbf{y}}}{|\boldsymbol{\lambda}|^2}.$$

Proposition 6.5

For $\alpha > 0$ and $\mathbf{x} \in \mathbb{R}^n$, $\mathcal{F}_{\mathbf{x} \rightarrow \boldsymbol{\lambda}}[|\mathbf{x}|^{-\alpha}] = C_{n,\alpha} |\boldsymbol{\lambda}|^{\alpha-n}$ where $C_{n,\alpha}$ is a constant depending on n and α .

Proof. Write $f_\alpha(\mathbf{x}) = |\mathbf{x}|^{-\alpha}$.

$$\begin{aligned} \hat{f}_\alpha(\boldsymbol{\lambda}) &= \int e^{-i\boldsymbol{\lambda} \cdot \mathbf{x}} |\mathbf{x}|^{-\alpha} d^n \mathbf{x} \\ &= \int e^{-i\boldsymbol{\lambda} \cdot R^\top \mathbf{x}'} |R^\top \mathbf{x}'|^{-\alpha} d^n \mathbf{x} \text{ where } \mathbf{x} = R^\top \mathbf{x}' \text{ with } R \in \text{SO}(n) \\ &= \int e^{-i(R\boldsymbol{\lambda}) \cdot \mathbf{x}'} |\mathbf{x}'|^{-\alpha} d^n \mathbf{x}' \\ &= \hat{f}_\alpha(R\boldsymbol{\lambda}) \end{aligned}$$

so $\hat{f}_\alpha$ is rotation invariant, that is if $\boldsymbol{\lambda} = |\boldsymbol{\lambda}| \mathbf{m}$, then $\hat{f}_\alpha(\boldsymbol{\lambda})$ is independent of $\mathbf{m}$. Substituting $|\boldsymbol{\lambda}| \mathbf{x} = \mathbf{x}'$,

$$\begin{aligned} \hat{f}_\alpha(\boldsymbol{\lambda}) &= \int e^{-i|\boldsymbol{\lambda}| \mathbf{m} \cdot \mathbf{x}} |\mathbf{x}|^{-\alpha} d^n \mathbf{x} \\ &= \left(\int e^{-i\mathbf{m} \cdot \mathbf{x}'} |\mathbf{x}'|^{-\alpha} d^n \mathbf{x}' \right) |\boldsymbol{\lambda}|^{\alpha-n} \\ &= C_{n,\alpha} |\boldsymbol{\lambda}|^{\alpha-n}. \end{aligned}$$

□

For $n > 2$, let $\alpha = n - 2 > 0$, so $\mathcal{F}_{\mathbf{x} \rightarrow \boldsymbol{\lambda}}[|\mathbf{x}|^{2-n}] = C_{n,n-2} |\boldsymbol{\lambda}|^{-2}$. That is,

$$\frac{1}{|\boldsymbol{\lambda}|^2} = \mathcal{F}_{\mathbf{x} \rightarrow \boldsymbol{\lambda}}[c_n |\mathbf{x}|^{2-n}]$$

where $c_n = C_{n,n-2}$.

Proposition 6.6

For $n > 2$,

$$G(\mathbf{x}, \mathbf{y}) = -\frac{1}{(n-2)|S^{n-1}|} \frac{1}{|\mathbf{x} - \mathbf{y}|^{n-2}}$$

where $|S^{n-1}|$ is the area of the $(n-1)$ -dimensional sphere. In particular, if $n = 3$,

$$G(\mathbf{x}, \mathbf{y}) = -\frac{1}{4\pi} \frac{1}{|\mathbf{x} - \mathbf{y}|}.$$

Proof. Set $F(\mathbf{x}) = c_n |\mathbf{x}|^{2-n}$, so $\hat{G}(\boldsymbol{\lambda}, \mathbf{y}) = e^{-i\boldsymbol{\lambda} \cdot \mathbf{y}} \hat{F}(\boldsymbol{\lambda}) = -\mathcal{F}_{\mathbf{x} \rightarrow \boldsymbol{\lambda}}[F(\mathbf{x} - \mathbf{y})]$. Hence,

$$G(\mathbf{x}, \mathbf{y}) = -\frac{c_n}{|\mathbf{x} - \mathbf{y}|^{n-2}}.$$

Since $\Delta G(\mathbf{x}, \mathbf{0}) = \delta(\mathbf{x})$, we have

$$\frac{1}{c_n} \delta(\mathbf{x}) = -\Delta[|\mathbf{x}|^{2-n}].$$

Integrating over $|\mathbf{x}| \leq 1$ and using the divergence theorem,

$$\begin{aligned} \frac{1}{c_n} &= -\int_{|\mathbf{x}| \leq 1} \Delta[|\mathbf{x}|^{2-n}] d^n \mathbf{x} \\ &= -\int_{|\mathbf{x}|=1} \nabla[|\mathbf{x}|^{2-n}] \cdot d\mathbf{S} \\ &= (n-2) \int_{|\mathbf{x}|=1} dS \\ &= (n-2) |S^{n-1}|. \end{aligned}$$

□

Proposition 6.7

If $n = 2$, a Green's function for Laplace's equation is

$$G(\mathbf{x}, \mathbf{y}) = \frac{1}{2\pi} \log |\mathbf{x} - \mathbf{y}|.$$

Proof. Take $\mathbf{y} = \mathbf{0}$ without loss of generality. Note that on $r = |\mathbf{x}| \neq 0$, $\Delta[\log r] = 0$ since

$$\Delta[\log r] = \frac{1}{r} \frac{\partial}{\partial r} \left[r \frac{\partial}{\partial r} \log r \right] = \frac{1}{r} \frac{\partial}{\partial r} [1] = 0.$$

Fix $\varepsilon > 0$ and use the divergence theorem,

$$\begin{aligned} \int_{|\mathbf{x}| \leq \varepsilon} \Delta[\log r] d^2 \mathbf{x} &= \int_{|\mathbf{x}|=\varepsilon} \nabla(\log r) \cdot d\mathbf{S} \\ &= \int_0^{2\pi} \frac{\mathbf{e}_r}{\varepsilon} \cdot \mathbf{e}_r \varepsilon d\theta \\ &= 2\pi. \end{aligned}$$

So, $\Delta[\log |\mathbf{x}| / (2\pi)] = 0$ for $\mathbf{x} \neq \mathbf{0}$ and

$$\int_{|\mathbf{x}| \leq \varepsilon} \Delta \left[\frac{1}{2\pi} \log |\mathbf{x}| \right] d^2 \mathbf{x} = 1$$

for all $\varepsilon > 0$. Hence, $\Delta[\log |\mathbf{x}| / (2\pi)] = \delta(\mathbf{x})$.

□

In summary,

$$G(\mathbf{x}, \mathbf{y}) = \begin{cases} \frac{1}{2\pi} \log |\mathbf{x} - \mathbf{y}| & \text{if } n = 2 \\ -\frac{1}{(n-2)|S^{n-1}|} \frac{1}{|\mathbf{x} - \mathbf{y}|^{n-2}} & \text{if } n > 2. \end{cases}$$

It is weird that we have a separate case when $n = 2$, however,

$$\lim_{n \rightarrow 2} -\frac{1}{(n-2)|S^{n-1}|} \left[\frac{1}{|\mathbf{x} - \mathbf{y}|^{n-2}} - 1 \right] = \frac{1}{2\pi} \log |\mathbf{x} - \mathbf{y}|.$$

For $\Omega \subseteq \mathbb{R}^n$, we want to solve

$$\begin{cases} \Delta u = F & \mathbf{x} \in \Omega \\ u = 0 & \mathbf{x} \in \partial\Omega. \end{cases}$$

To do this, we can find the Dirichlet Green's function, defined for each $\mathbf{y} \in \Omega$ by

$$\begin{cases} \Delta \mathcal{G}(\mathbf{x}, \mathbf{y}) = \delta(\mathbf{x} - \mathbf{y}) & \mathbf{x} \in \Omega \\ \mathcal{G}(\mathbf{x}, \mathbf{y}) = 0 & \mathbf{x} \in \partial\Omega. \end{cases}$$

Given such $\mathcal{G}(\mathbf{x}, \mathbf{y})$, we see that

$$u(\mathbf{x}) = \int_{\Omega} \mathcal{G}(\mathbf{x}, \mathbf{y}) F(\mathbf{y}) d^n \mathbf{y}$$

solves the problem. To find $\mathcal{G}(\mathbf{x}, \mathbf{y})$, we need to add on to the free space Green's function some harmonic function so that $\mathcal{G}(\mathbf{x}, \mathbf{y}) = 0$ when $\mathbf{x} \in \partial\Omega$. We can use the method of images. We add a series of charges $\{\mathbf{y}_0, \mathbf{y}_1, \dots\} \subseteq \Omega^c$ such that the total potential due to $\mathbf{y}$ and $\{\mathbf{y}_0, \mathbf{y}_1, \dots\}$ vanishes on $\partial\Omega$.

Example 6.9

Take $\Omega = \{\mathbf{x} \in \mathbb{R}^3 : x_3 > 0\}$. Given a charge at $\mathbf{y} = (y_1, y_2, y_3)$, we can add another charge of opposite polarity at $\mathbf{y}_0 = (y_1, y_2, -y_3) \in \Omega^c$. At $\partial\Omega = \{\mathbf{x} \in \mathbb{R}^3 : x_3 = 0\}$, the electric potentials cancel. Hence, we guess $\mathcal{G}(\mathbf{x}, \mathbf{y}) = G(\mathbf{x}, \mathbf{y}) + [-G(\mathbf{x}, \mathbf{y}_0)]$. For each $\mathbf{x} \in \Omega$,

$$\Delta \mathcal{G}(\mathbf{x}, \mathbf{y}) = \delta(\mathbf{x} - \mathbf{y}) - \delta(\mathbf{x} - \mathbf{y}_0) = \delta(\mathbf{x} - \mathbf{y})$$

since $\mathbf{x} \neq \mathbf{y}_0$. For $\mathbf{x} \in \partial\Omega$,

$$\mathcal{G}(\mathbf{x}, \mathbf{y}) = -\frac{1}{4\pi} \left[\frac{1}{\sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (0 - y_3)^2}} - \frac{1}{\sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (0 + y_3)^2}} \right] = 0.$$

The solution is

$$u(\mathbf{x}) = \int_{\Omega} [G(\mathbf{x}, \mathbf{y}) - G(\mathbf{x}, \mathbf{y}_0)] F(\mathbf{y}) d^3 \mathbf{y}.$$

Note that $\mathbf{y}_0$ depends on $\mathbf{y}$.

6.7 Green's function for wave equation

We want to solve the IVP

$$\begin{cases} u_{tt} - c^2 \Delta u = F(\mathbf{x}, t) & (\mathbf{x}, t) \in \mathbb{R}^n \times (0, \infty) \\ u(\mathbf{x}, 0) = f(\mathbf{x}) & \mathbf{x} \in \mathbb{R}^n \\ u_t(\mathbf{x}, 0) = g(\mathbf{x}) & \mathbf{x} \in \mathbb{R}^n. \end{cases}$$

We will again split this into two problems.

- (i) Zero forcing ($F = 0$) and arbitrary initial data.
- (ii) Zero initial data ($f = 0$ and $g = 0$) and arbitrary forcing.

To solve this, we will use the function $\Phi_t(\mathbf{x})$ defined implicitly by

$$\hat{\Phi}_t(\boldsymbol{\lambda}) = \frac{\sin(c|\boldsymbol{\lambda}|t)}{c|\boldsymbol{\lambda}|}.$$

That is, by inversion,

$$\Phi_t(\mathbf{x}) = \frac{1}{(2\pi)^n} \int e^{i\boldsymbol{\lambda} \cdot \mathbf{x}} \frac{\sin(c|\boldsymbol{\lambda}|t)}{c|\boldsymbol{\lambda}|} d^n \boldsymbol{\lambda}.$$

This is difficult to compute in arbitrary dimensions.

Proposition 6.8

When $n = 1$,

$$\Phi_t(x) = \frac{1}{2c} [H(x + ct) - H(x - ct)].$$

Proof. Note that

$$H(x + ct) - H(x - ct) = \begin{cases} 0 & \text{if } x < -ct \\ 1 & \text{if } |x| < ct \\ 0 & \text{if } x > ct. \end{cases}$$

That is, it is the indicator function on $(-ct, ct)$. So,

$$\begin{aligned} \hat{\Phi}_t(\lambda) &= \frac{1}{2c} \int_{-ct}^{ct} e^{-i\lambda x} dx \\ &= -\frac{1}{2ci\lambda} [e^{-i\lambda ct} - e^{i\lambda ct}] \\ &= \frac{\sin(c\lambda t)}{c\lambda} \\ &= \frac{\sin(c|\lambda|t)}{c|\lambda|}. \end{aligned}$$

Since $\hat{\Phi}_t(\lambda)$ determines $\Phi_t(x)$, we are done. $\square$

Proposition 6.9

When $n = 3$,

$$\Phi_t(\mathbf{x}) = \frac{1}{4\pi c|\mathbf{x}|} [\delta(|\mathbf{x}| - ct) - \delta(|\mathbf{x}| + ct)].$$

Proof. We use spherical polars $(\lambda_1, \lambda_2, \lambda_3) = (r \sin \theta \cos \phi, r \sin \theta \sin \phi, r \cos \theta)$ where $(r, \theta, \phi) \in [0, \infty) \times [0, \pi] \times [0, 2\pi)$. Without loss of generality, align the λ_3 -axis in the direction of $\mathbf{x}$ so that $\mathbf{x} \cdot \boldsymbol{\lambda} = |\mathbf{x}||\boldsymbol{\lambda}| \cos \theta = |\mathbf{x}|r \cos \theta$. We have, for $R > 0$,

$$\begin{aligned} \int_{|\mathbf{x}| < R} e^{i\boldsymbol{\lambda} \cdot \mathbf{x}} \hat{\Phi}_t(\boldsymbol{\lambda}) d^3 \boldsymbol{\lambda} &= \int_0^{2\pi} d\phi \int_0^\pi d\theta \int_0^R dr e^{i|\mathbf{x}|r \cos \theta} \frac{\sin(crt)}{cr} r^2 \sin \theta \\ &= \frac{2\pi}{c} \int_0^R r dr \int_0^\pi e^{i|\mathbf{x}|r \cos \theta} \sin \theta \sin(crt) d\theta \\ &= \frac{2\pi}{c} \int_0^R dr r \sin(crt) \int_0^\pi \frac{\partial}{\partial \theta} [e^{i|\mathbf{x}|r \cos \theta}] \frac{1}{-ir|\mathbf{x}|} d\theta \\ &= \frac{2\pi}{-ic|\mathbf{x}|} \int_0^R \sin(crt) [e^{-i|\mathbf{x}|r} - e^{i|\mathbf{x}|r}] dr \\ &= \frac{4\pi}{c|\mathbf{x}|} \int_0^R \sin(crt) \sin(r|\mathbf{x}|) dr \\ &= \frac{\pi}{c|\mathbf{x}|} \operatorname{Re} \left[\int_{-R}^R (e^{ir(|\mathbf{x}|-ct)} - e^{ir(|\mathbf{x}|+ct)}) dr \right] \rightarrow \frac{\pi}{c|\mathbf{x}|} [2\pi\delta(|\mathbf{x}| - ct) - 2\pi\delta(|\mathbf{x}| + ct)] \end{aligned}$$

as $r \rightarrow \infty$. Use the fact that

$$\Phi_t(\mathbf{x}) = \frac{1}{(2\pi)^3} \lim_{R \rightarrow \infty} \int_{|\boldsymbol{\lambda}| < R} e^{i\boldsymbol{\lambda} \cdot \mathbf{x}} \hat{\Phi}_t(\boldsymbol{\lambda}) d^3 \boldsymbol{\lambda}$$

and we are done. $\square$

In both cases, we have $\Phi_t(\mathbf{x}) = \Phi_t^+(\mathbf{x}) - \Phi_t^-(\mathbf{x})$. We call $\Phi_t^+(\mathbf{x})$ the advanced part and $\Phi_t^-(\mathbf{x})$ the retarded

part of Φ_t . For example, when $n = 3$,

$$\Phi_t^\pm(\mathbf{x}) = \frac{1}{4\pi c|\mathbf{x}|} \delta(|\mathbf{x}| \mp ct).$$

Proposition 6.10

The solution to (i) is

$$u(\mathbf{x}, t) = (\Phi_t * g)(\mathbf{x}) + \frac{\partial}{\partial t}[(\Phi_t * f)(\mathbf{x})].$$

Proof. Take the Fourier transform of the PDE and the initial conditions.

$$\hat{u}_{tt} + c^2|\boldsymbol{\lambda}|^2\hat{u} = 0 \text{ with } \hat{u}(\boldsymbol{\lambda}, 0) = \hat{f}(\boldsymbol{\lambda}) \text{ and } \hat{u}_t(\boldsymbol{\lambda}, 0) = \hat{g}(\boldsymbol{\lambda}).$$

The general solution to the ODE is $\hat{u}(\boldsymbol{\lambda}, t) = A(\boldsymbol{\lambda}) \sin(c|\boldsymbol{\lambda}|t) + B(\boldsymbol{\lambda}) \cos(c|\boldsymbol{\lambda}|t)$. The initial conditions give $B(\boldsymbol{\lambda}) = \hat{f}(\boldsymbol{\lambda})$ and $c|\boldsymbol{\lambda}|A(\boldsymbol{\lambda}) = \hat{g}(\boldsymbol{\lambda})$. That is,

$$\hat{u}(\boldsymbol{\lambda}, t) = \frac{\sin(c|\boldsymbol{\lambda}|t)}{c|\boldsymbol{\lambda}|} \hat{g}(\boldsymbol{\lambda}) + \cos(c|\boldsymbol{\lambda}|t) \hat{f}(\boldsymbol{\lambda}) = \hat{\Phi}_t(\boldsymbol{\lambda}) \hat{g}(\boldsymbol{\lambda}) + \frac{\partial}{\partial t}[\hat{\Phi}_t(\boldsymbol{\lambda}) \hat{f}(\boldsymbol{\lambda})].$$

Apply the convolution theorem to get the result. □

Look at the term $(\Phi_t * g)(\mathbf{x}) = \int \Phi_t(\mathbf{x} - \mathbf{y})g(\mathbf{y}) d^n \mathbf{y}$. When $n = 3$ and $t > 0$, we have $\Phi_t(\mathbf{x} - \mathbf{y}) = \Phi_t^+(\mathbf{x} - \mathbf{y})$ and

$$\begin{aligned} (\Phi_t * g)(\mathbf{x}) &= \frac{1}{4\pi c} \int \frac{\delta(|\mathbf{x} - \mathbf{y}| - ct)}{|\mathbf{x} - \mathbf{y}|} g(\mathbf{y}) d^3 \mathbf{y} \\ &= \frac{t}{4\pi(ct)^2} \int_{|\mathbf{x} - \mathbf{y}| = ct} g(\mathbf{y}) dS \\ &= t \times (\text{the average of } g \text{ on a sphere of radius } ct \text{ centred at } \mathbf{x}). \end{aligned}$$

Suppose a friend at $\mathbf{x}_0$ sends off a very short pulse of light at $t = 0$. You, standing at the origin, will see a flash at time $t = |\mathbf{x}_0|/c$. However, the propagation of light is different depending on whether n is odd or even. In even dimensions, $\Phi_t(\mathbf{x} - \mathbf{y}) \propto$ (the average of g within a sphere of radius ct centred at $\mathbf{x}$) and you will see light starting at $t = |\mathbf{x}_0|/c$ which slowly dims over time.

Proposition 6.11

The solution to (ii) is

$$u(\mathbf{x}, t) = \int_0^t \left[\int \Phi_{t-s}(\mathbf{x} - \mathbf{y}) F(\mathbf{y}, s) d^n \mathbf{y} \right] ds.$$

Proof. Take the Fourier transform of the PDE and the initial conditions.

$$\hat{u}_{tt} + c^2|\boldsymbol{\lambda}|^2\hat{u} = \hat{F}(\boldsymbol{\lambda}, t) \text{ with } \hat{u}(\boldsymbol{\lambda}, 0) = \hat{u}_t(\boldsymbol{\lambda}, 0) = 0.$$

To solve the ODE, we need the Green's function for $L = d^2/dt^2 + c^2|\boldsymbol{\lambda}|^2$. We get

$$G(t, s) = \begin{cases} 0 & \text{if } t < s \\ \frac{\sin(c|\boldsymbol{\lambda}|(t-s))}{c|\boldsymbol{\lambda}|} & \text{if } t > s \end{cases} = \begin{cases} 0 & \text{if } t < s \\ \hat{\Phi}_{t-s}(\boldsymbol{\lambda}) & \text{if } t > s. \end{cases}$$

Hence,

$$\hat{u}(\boldsymbol{\lambda}, t) = \int_0^t \hat{\Phi}_{t-s}(\boldsymbol{\lambda}) \hat{F}(\boldsymbol{\lambda}, s) ds.$$

By the convolution theorem, we get the result. □

The solution to the general IVP is given by superposition. Note that the solution to (ii) can be written as

$$u(\mathbf{x}, t) = \int ds \int d^n \mathbf{y} G(\mathbf{x}, t; \mathbf{y}, s) F(\mathbf{y}, s)$$

if we set $G(\mathbf{x}, t; \mathbf{y}, s) = H(t - s) \Phi_{t-s}(\mathbf{x} - \mathbf{y})$ and $F(\mathbf{y}, t)$ when $t < 0$. We call $G(\mathbf{x}, t; \mathbf{y}, s)$ the Green's function for the wave equation. We can show that

$$\frac{\partial^2 G}{\partial t^2} - c^2 \Delta G = \delta(t - s) \delta(\mathbf{x} - \mathbf{y}).$$

6.8 Wave equation on half-line

We want to solve the IBVP

$$\begin{cases} v_{tt} - c^2 v_{xx} = P(x, t) & x > 0 \text{ and } t > 0 \\ v(0, t) = p(t) & t > 0 \\ v(x, 0) = q(x) & x > 0 \\ v_t(x, 0) = r(x) & x > 0 \end{cases}$$

with $p(0) = q(0)$ and $p'(0) = r(0)$.

Proposition 6.12

The above IBVP is equivalent to

$$\begin{cases} u_{tt} - c^2 u_{xx} = F(x, t) & x > 0 \text{ and } t > 0 \\ u(0, t) = 0 & t > 0 \\ u(x, 0) = f(x) & x > 0 \\ u_t(x, 0) = g(x) & x > 0 \end{cases}$$

with $F(0, t) = 0$ and $f(0) = g(0) = 0$.

Proof. Let $v(x, t)$, $P(x, t)$ and $p(t)$ be as in the original IBVP. Define

$$u(x, t) = v(x, t) - [p(t + x/c) + Q(t) - Q(t + x/c)] \text{ where } Q(t) = \int_0^t (t - s) P(0, s) ds.$$

Note that $Q(0) = 0$, $Q'(0) = 0$ and $Q''(t) = P(0, t)$. Apply $L = \partial_t^2 - c^2 \partial_x^2$ to u and get

$$u_{tt} - c^2 u_{xx} = P(x, t) - Q''(t) = P(x, t) - P(0, t) \equiv F(x, t).$$

Also,

$$\begin{aligned} u(0, t) &= p(t) - p(t) = 0 \\ u(x, 0) &= q(x) - p(x/c) + Q(x/c) \equiv f(x) \\ u_t(x, 0) &= r(x) - p'(x/c) + Q'(x/c) \equiv g(x). \end{aligned}$$

We see that $f(0) = q(0) - p(0) = 0$ and $g(0) = r(0) - p'(0) = 0$. Hence, u satisfies the modified IBVP if and only if v satisfies the original IBVP. $\square$

Once again, we break this modified problem into two.

- (i) Zero forcing ($F = 0$) and arbitrary initial conditions.
- (ii) Zero initial conditions ($f = 0$ and $g = 0$) and arbitrary forcing.

To solve (i), imagine a semi-infinite string clamped at $x = 0$. We extend the string to the whole real line and try

to recreate the same motion on $x > 0$ with the infinite string. Choose the initial data to be $f_{\text{odd}}(x)$ and $g_{\text{odd}}(x)$ for this problem on the full line, where

$$f_{\text{odd}}(x) = \begin{cases} f(x) & \text{if } x \geq 0 \\ -f(-x) & \text{if } x < 0 \end{cases}$$

and similarly for $g_{\text{odd}}(x)$. Note that this is continuous since $f(0) = 0$.

Proposition 6.13

The solution to (i) is

$$u(x, t) = (\Phi_t * g_{\text{odd}})(x) + \frac{\partial}{\partial t}(\Phi_t * f_{\text{odd}})(x).$$

Proof. Consider the problem on the full line

$$\begin{cases} w_{tt} - c^2 w_{xx} = 0 & x \in \mathbb{R} \text{ and } t > 0 \\ w(x, 0) = f_{\text{odd}}(x) & x \in \mathbb{R} \\ w_t(x, 0) = g_{\text{odd}}(x) & x \in \mathbb{R}. \end{cases}$$

We know that

$$\begin{aligned} w(x, t) &= (\Phi_t * g_{\text{odd}})(x) + \frac{\partial}{\partial t}(\Phi_t * f_{\text{odd}})(x) \\ \hat{w}(\lambda, t) &= \hat{\Phi}_t(\lambda) \hat{g}_{\text{odd}}(\lambda) + \frac{\partial}{\partial t} [\hat{\Phi}_t(\lambda) \hat{f}_{\text{odd}}(\lambda)]. \end{aligned}$$

Note that $\hat{\Phi}_t$ is even. We can show that $x \mapsto \sigma(x)$ is odd/even if and only if $\lambda \mapsto \hat{\sigma}(\lambda)$ is odd/even. Since $\hat{w}(\lambda, t)$ is odd, $w(x, t)$ is odd. Hence, $w(0, t) = 0$ and we can deduce that $u(x, t) = w(x, t)$ when $x > 0$. $\square$

Proposition 6.14

The solution to (ii) is

$$u(x, t) = \int_0^t \left[\int \Phi_{t-s}(x-y) F_{\text{odd}}(y, s) dy \right] ds$$

where $F_{\text{odd}}(x, t)$ is defined similarly as $f_{\text{odd}}(x)$.

Proof. The solution to

$$\begin{cases} w_{tt} - c^2 w_{xx} = F_{\text{odd}}(x, t) & x \in \mathbb{R} \text{ and } t > 0 \\ w(x, 0) = 0 & x \in \mathbb{R} \\ w_t(x, 0) = 0 & x \in \mathbb{R} \end{cases}$$

is

$$\begin{aligned} w(x, t) &= \int_0^t \left[\int \Phi_{t-s}(x-y) F_{\text{odd}}(y, s) dy \right] ds \\ \hat{w}(\lambda, t) &= \int_0^t \hat{\Phi}_{t-s}(\lambda) \hat{F}_{\text{odd}}(\lambda) ds. \end{aligned}$$

Since $\hat{w}(\lambda, t)$ is odd, $w(x, t)$ is odd. Hence, $w(0, t) = 0$ and $u(x, t) = w(x, t)$ when $x > 0$. $\square$